Chapter 7

Profit and Loss

Theory:

Cost Price (CP): The money paid by the shopkeeper to the manufacturer or whole -seller to buy goods is called the cost price (cp) of the goods purchased by the shopkeeper.

Selling Price (SP): The price at which the shopkeeper sells the goods is called selling price (s.p) of the goods sold by the shopkeeper to the customer.

Profit: If the selling price of an article is more than its cost price, then the dealer (or shopkeeper) makes a profit (or gain)

i.e., Profit = SP – CP; SP > CP

Loss: If the selling price of an article is less than its cost price, the dealer suffers a loss

i.e., Loss = CP – SP; CP > SP

Some Important Formulae:

(i) Profit = SP – CP

(ii) Loss = CP – SP

(iii) Profit percentage = $\left(\frac{\text{Profit}}{\text{CP}} \times 100\right)\%$

(iv) Loss percentage = $\left(\frac{\text{Loss}}{\text{CP}} \times 100\right)\%$

(v) S.P = $\left(\frac{(100+\text{Profit}\%)\times \text{CP}}{100}\right) = \left(\frac{(100-\text{Loss}\%)\times \text{CP}}{100}\right)$

(vi) C.P = $\left(\frac{100\times \text{SP}}{100+\text{Profit}\%}\right) = \left(\frac{100\times \text{SP}}{100-\text{Loss}\%}\right)$

(vii) SP = (100 + x)% of CP; when Profit = x% of CP

(viii) SP = (100 – x)% of CP; when Loss = x% of CP

Example 1: A man purchases an item for Rs. 120 and he sells it at a 20 percent profit, find his selling price

Sol. SP = $\left(\frac{100+\text{Profit}\%}{100}\right)\times \text{CP} = \frac{100+20}{100}\times 120 = \frac{120}{100}\times 120 = \text{Rs. } 144$

Note: **Profit/Loss percentage is always calcualated on C.P. unless otherwise stated.**

Example 2: Find the cost price of an article which is sold for Rs. 200 at a loss of 20%

Sol. CP = $\frac{100}{100-\text{Loss}\%}\times \text{SP} = \frac{100}{100-20}\times 200$ = Rs. 250

Concept 1:

MARK UP AND DISCOUNT

Marked Price: To avoid loss due to bargaining by the customer and to get profit over the cost price, the trader increases the cost price. This increase is known as markup and the increased price (i.e., cp+markup) is called the marked price or printed price or list price of the goods.

Marked Price = CP + markup

$$\text{Marked Price} = \text{CP} + \frac{(\%\text{ marked})\times \text{CP}}{100}$$

Generally goods are sold at marked price. If there is no further discount, then in this case selling price equals marked price.

Discount: Discount means reduction of marked price to sell at a lower rate or literally discount means concession. Basically, it is calculated on the basis of marked price.

Selling price = Marked price – Discount

$$\text{Selling price} = \text{MP} - \frac{(\%\ \text{Discount}) \times \text{MP}}{100}$$

Example: If the cost price of an article is Rs. 300 and the percent markup is 30%. What is the marked price?

Sol. $\text{MP} = \text{CP} + (\%\text{markup on CP}) = 300 + \frac{30}{100} \times 300 = \text{Rs. } 390$

Concept 2:

Dishonest Dealer Case: If a trader professes to sell his goods at cost price, but uses false weights, then

$$\%\ \text{gain} = \frac{\text{Error}}{\text{True value} - \text{Error}} \times 100 \quad \Rightarrow \quad \%\text{gain} = \frac{\text{True weight} - \text{False weight}}{\text{False weight}} \times 100$$

Example: A shopkeeper sold an article at cost price but use the weight of 960 gm in place of 1 kg weight. Find his profit%?

Sol. $\text{Profit}\% = \frac{\text{True weight} - \text{False weight}}{\text{False weight}} \times 100 = \frac{1000 - 960}{960} \times 100 = \frac{40}{960} \times 100 = \frac{25}{6} = 4\frac{1}{6}\%$

Concept 3:

Where two articles are sold at same price but one of them at a profit and another at a loss and the percentage profit is the same as the percentage loss, In this case there is always a loss.

$$\text{Loss}\% = \left(\frac{\text{Common Profit or Loss}\%}{10}\right)^2 = \left(\frac{\%\text{value}}{10}\right)^2$$

Example: Each of two car is sold for Rs. 1000. The first one is sold at 25% profit and the other one at 25% loss. What is the percentage loss or gain in the deal?

Sol. Total s.p $= 1000 \times 2 = \text{Rs. } 2000$

$$\text{CP of } 1^{\text{st}} \text{ car} = \frac{100 \times 1000}{125} \quad [\because \text{Profit} = 25\%]$$

$$= \text{Rs. } 800$$

$$\text{CP of } 2^{\text{nd}} \text{ car} = \frac{100 \times 1000}{75} \quad [\because \text{Loss} = 25\%]$$

$$= \text{Rs. } 1333\frac{1}{3}$$

$$\text{Total CP} = \text{Rs. } 2133\frac{1}{3} \quad \Rightarrow \text{Loss}\% = \frac{\text{CP} - \text{SP}}{\text{CP}} \times 100 = \frac{2133\frac{1}{3} - 2000}{2133\frac{1}{3}} \times 100 = 6.25\%$$

or, Using Shortcut Formula

$$\text{Loss}\% = \left(\frac{\%\text{value}}{10}\right)^2 = \left(\frac{25}{10}\right)^2 = 6.25\ \%$$

Concept 4:

When two successive discounts on an article are x% and y% resp. then net discount: $\left(x + y - \frac{xy}{100}\right)\%$

Example: A shopkeeper gives two sucessive discount of 50% and 50%. Find the real (equivalent) discount?

Sol. Let MP = Rs. 100

Cost after 1^{st} discount of 50% = 100 – 50% of 100 = Rs. 50

Cost after 2^{nd} discount of 50% = 50 – 50% of 50 = Rs. 25
Price after both discount = Rs. 25

$$\% \text{ discount} = \frac{100-25}{100} \times 100 = 75\%$$

or, Using Shortcut Formula

$$\% \text{ discount} = x + y - \frac{xy}{100} \qquad [\text{where } x = 50\%, y = 50\%]$$

$$= 50 + 50 - \frac{50 \times 50}{100} = 100 - 25 = 75\%$$

Types of Questions

1. There is a profit of 20% on the cost price of an article. Find the profit percent when calculated on selling price?

Sol. Let the cost price of an article be Rs. 100
then, Profit = 20% of 100 = Rs. 20
Selling price = Cost price + profit
= 100 + 20 = Rs. 120
Profit% when calculated on SP

$$= \frac{20}{120} \times 100 = \frac{100}{6} = 16\frac{2}{3}\%$$

2. By selling a bicycle for Rs. 2850, a shopkeeper gains 14%. If the profit is reduced to 8%, find the selling price of bicycle?

Sol. $$CP = \frac{SP \times 100}{100 + \text{Profit}\%} = \frac{2850 \times 100}{100 + 14}$$

$$= \frac{2850 \times 100}{114} = \text{Rs. } 2500$$

SP of article for 8% Profit

$$SP = \frac{CP \times (100 + \text{Profit}\%)}{100} = \frac{2500 \times 108}{100}$$

= 25 × 108 = Rs. 2700

3. The selling price of 12 articles is equal to the cost price of 15 articles. Find the gain percent?

Sol. Let the CP of 1 article = Rs. x
Cost Price of 15 article = Rs. 15x
Selling Price of 12 article = Rs. 15x

$$\text{SP of 1 article} = \text{Rs. } \frac{15}{12}x$$

$$\text{Gain} = \frac{15x}{12} - x = \frac{3x}{12} = \frac{x}{4}$$

$$\text{Gain\%} = \frac{\text{Gain} \times 100}{CP} = \frac{\frac{x}{4} \times 100}{x} = 25\%$$

4. A fruit seller buys some fruits at the rate of 11 for Rs. 10 and the same number at the rate of 9 for Rs. 10. If all the fruits are sold for Rs. 1 each. Find the gain or loss percent?

Sol. In these types of question, we have to take the LCM of number of individual things.
Number of fruits of each type be bought
= LCM of 11 and 9 = 99
Total number of fruits = 99 × 2 = 198

$$\text{CP of 198 fruits} = \frac{10}{11} \times 99 + \frac{10}{9} \times 99$$

= 90 + 110 = Rs. 200
SP = 198 × 1 = Rs. 198

$$\text{Loss\%} = \frac{CP - SP}{CP} \times 100 = \frac{200 - 198}{200} \times 100$$

$$= \frac{2}{200} \times 100 = 1\%$$

5. A book vendor sold a book at a loss of 10%. Had he sold it for Rs. 108 more, he would have earned a profit of 10%. Find the cost of the book.

Sol. Let the CP article = x

$$SP = \frac{x(100-10)}{100} = \frac{90x}{100} = \frac{9x}{10} \quad [\text{Loss of } 10\%]$$

$$\frac{9x}{10} + 108 = \frac{110x}{100} \quad [\text{If vendor sold for Rs. 108 more}]$$

$$\frac{110x}{100} - \frac{9x}{10} = 108 \quad \Rightarrow \quad \frac{11x}{10} - \frac{9x}{10} = 108$$

$$2x = 1080 \quad \Rightarrow \quad x = \text{Rs. } 540$$

6. A person bought some articles at the rate of 5 per rupee and the same number at the rate of 4 per rupee. He mixed both the types and sold at the rate of 9 for Rs. 2. In this business he suffered a loss of Rs. 3. Find th total no. of articles bought by him?

Sol. Let the person buys 10 articles

$$\text{Total CP} = \text{Rs.}\left(5 \times \frac{1}{5} + \frac{5 \times 1}{4}\right) = \text{Rs.}\left(1 + \frac{5}{4}\right) = \text{Rs.}\frac{9}{4}$$

$$\text{SP of 10 articles} = \frac{2}{9} \times 10 = \text{Rs.}\frac{20}{9}$$

$$\text{Loss} = \text{Rs.}\left(\frac{9}{4}-\frac{20}{9}\right) = \left(\frac{81-80}{36}\right) = \text{Rs.}\frac{1}{36}$$

If loss is Rs. $\frac{1}{36}$, then number of articles = 10

If loss is Rs. 3, number of articles = 36 × 10 × 3 = 1080

7. A man buys a field of agricultural land for Rs. 360000. He sells $\frac{1}{3}$rd at a loss of 20% and $\frac{2}{5}$th at a gain of 25%. At what price must he sell the remaining field so as to make an overall profit of 10%?

Sol. SP of total agricultural field $= \text{Rs.}\left(360000 \times \frac{110}{100}\right)$

= Rs. 396000 [overall profit of 10%]

SP of $\frac{1}{3}$rd of the field

$$= \frac{1}{3} \times 360000 \times \frac{80}{100}\ [\text{Loss of } 20\%] \quad \Rightarrow \quad \text{Rs. } 96000$$

SP of $\frac{2}{5}$th of the field

$$= \frac{2}{5} \times 360000 \times \frac{125}{100}\ [\text{Gain of } 25\%] \quad \Rightarrow \quad \text{Rs. } 180000$$

SP of the remaining field

= Rs. (396000 – 96000 – 180000) = Rs. 120000

8. One trader calculates the percentage of profit on the buying price and another calculates on the selling price. When their selling price are the same, then difference of their actual profit is Rs. 85 and both claim to have made 20% profit. What is the selling price of each?

Sol. For first trader,

Let the CP of the article of Rs. 100, SP = Rs. 120

For second trader, SP of the article = Rs. 120

Gain = 20% [For both the traders]

Let the CP be x

$$\frac{120-x}{120} \times 100 = 20 \quad \Rightarrow \quad 120 - x = \frac{20}{5} \times 6$$

$\Rightarrow$ 120 – x = 24 $\Rightarrow$ x = 120 – 24 = Rs. 96

Gain = Rs. 24 [SP – CP]

Difference of gain = 24 – 20 = Rs. 4

If the difference of gain be Rs. 4, then

SP = Rs. 120

When the difference be Rs. 85, then

$$\text{SP} = \frac{120}{4} \times 85 = \text{Rs. } 2550$$

9. If the sales tax be reduced from $3\frac{1}{2}\%$ to $3\frac{1}{3}\%$. What difference does it make to person who purchases an article whose marked price is Rs. 8400?

Sol. Initial sales tax $= 3\frac{1}{2}\%$, Final sales tax $= 3\frac{1}{3}\%$

Difference in percentage of sales tax

$$= \left(3\frac{1}{2} - 3\frac{1}{3}\right)\% = \frac{1}{6}\%$$

$$\text{Req. diff.} = \frac{1}{6}\% \times 8400 = \frac{1}{6} \times \frac{1}{100} \times 8400 = \text{Rs. } 14$$

10. A man sells two cycles for Rs. 1710. The cost price of the first is equal to the selling price of the second. If the first is sold at 10% loss and the second at 25% gain, what is his total gain or loss?

Sol.

	1st Cycle	2nd Cycle	Total
CP	100	$100\left(\frac{100}{125}\right) = 80$	180
SP	$100\left(\frac{90}{100}\right) = 90$	100	190

Total CP = (CP of 1st Cycle) + (CP of 2nd Cycle)

= 100 + 80 = Rs. 180

Total SP = (SP of 1st Cycle) + (SP of 2nd Cycle)

= 90 + 100 = Rs. 190

CP : SP = 180 : 190 = 18 : 19

$$\text{Profit} = \frac{19-18}{19} \times 1710 = \text{Rs. } 90$$

11. Ashish bought an article with 20% discount on the labelled price. He sold the article with 30% profit on the labelled price. What was his percent profit on the price he bought?

Sol. Let the labelled price of the article be Rs. x

$$\text{Cost Price} = x\left(\frac{100-20}{100}\right) = \text{Rs.}\frac{4x}{5}$$

$$\text{Selling Price} = x\left(\frac{100+30}{100}\right) = \text{Rs.}\frac{13}{10}x$$

$$\text{Profit} = \frac{13}{10}x - \frac{4}{5}x = \frac{13x-8x}{10} = \frac{x}{2} \quad [\text{SP} - \text{CP}]$$

$$\%\text{Profit} = \frac{x/2}{4x/5} \times 100 = \frac{5}{8} \times 100 = \frac{125}{2} = 62.5\%$$

12. A shopkeeper sold an article for Rs. 400 after giving 20% discount on the labelled price and made 20% profit on cost price. What was the percentage profit, had he not given the discount?

Sol. $\text{Labelled Price} = \frac{400 \times 100}{80}$ [Before discount of 20%]

= Rs. 500

Cost Price of article

$$= \frac{400 \times 100}{120} = \text{Rs.} \frac{1000}{3} \quad [20\% \text{ profit on CP}]$$

$$\text{Profit\%} = \frac{500 - \frac{1000}{3}}{\frac{1000}{3}} \times 100 = \frac{\frac{1500 - 1000}{3}}{\frac{1000}{3}} \times 100$$

$$= \frac{500}{1000} \times 100 = 50\%$$

13. A reduction of 20% in the price of mangoes enables a person to purchase 12 more for Rs. 15. Find the price of 16 mangoes before reduction?

Sol. Let the price of 1 mango be x paise

Number of mangoes for

$$\text{Rs. } 15 = \frac{1500}{x} \qquad [\text{Rs. } 1 = 100 \text{ paise}]$$

$$\text{New price of one mango} = (80\% \text{ of } x) \text{ paise}$$

$$= \frac{80}{100} \times x = \frac{4}{5} x \text{ paise}$$

$$\text{Number of mangoes for Rs. } 15 = \left(\frac{1500 \times 5}{4x}\right)$$

$$\frac{7500}{4x} - \frac{1500}{x} = 12 \quad [\text{Diff. as mentioned in the Ques.}]$$

x = 31.25

Cost of 16 mangoes before reduction

$$= \frac{31.25 \times 16}{100} = \text{Rs. } 5$$

14. A garment company declared 15% discount for wholesale buyers. Mr. Hemant bought garments from the company for Rs. 8500 after getting discount. He fixed up selling price of garments in such a way that he earned a profit of 10% on original company price. What is the total selling price?

Sol. $\text{Original Company price} = \frac{8500 \times 100}{100 - 15} = \text{Rs. } 10000$

Let the total selling price be Rs. x.

Now, according to the question,

$$\frac{x - 10000}{10000} \times 100 = 10 \qquad [\text{Profit of } 10\%]$$

$100x - 1000000 = 100000 \quad \Rightarrow \quad x = \text{Rs. } 11000$

Total selling price = Rs. 11000

15. A publisher published 5000 books in 5 lakh rupees. He gives 500 books in free, $\frac{2}{3}$rd of the rest he sell on 20% discount and remaining $\frac{1}{3}$rd on M.P. He also gives 20% commission of the total selling. Find the profit% of the publisher if market price of each book is Rs. 200?

Sol. Total number of books = 5000

he gives free book = 500

$$\text{SP of I}^{\text{st}} \text{ part} = 3000 \times 200 \times \frac{4}{5} = \text{Rs. } 480000$$

$$\left[20\% \text{ Discount on } \frac{2}{3}\text{rd of rest}\right]$$

SP or II$^{\text{nd}}$ part = 1500 × 200 = Rs. 300000

$$\left[\text{Price is MP of } \frac{1}{3}\text{rd of the rest}\right]$$

Total SP = 480000 + 300000 = Rs. 780000

$$\text{Total SP after Commission} = \frac{80}{100} \times 780000$$

[20% Commission]

= Rs. 624000

Total CP = Rs. 5,00,000, Total SP = Rs. 6,24,000

Net profit = 6,24,000 – 5,00,000 = 1,24,000

$$\text{Profit\%} = \frac{124000}{500000} \times 100 = 24.8\%$$

Foundation

Questions

1. A man buys an article for Rs. 27.50 and sells it for Rs. 28.60. Find the gain percent?

(a) 4% (b) 3%

(c) 5% (d) 10%

2. If a radio is purchased for Rs. 490 and sold for Rs. 465.50. Find the loss%?

(a) 6% (b) 5%

(c) 4% (d) 3%

3. Find SP when CP = Rs. 56.25 and Gain = 20%?

(a) Rs. 72 (b) Rs. 67.5

(c) Rs. 50 (d) Rs. 75

4. Find SP when CP = Rs. 80.40, loss = 5%?

(a) Rs. 81 (b) Rs. 84.72

(c) Rs. 76.38 (d) Rs. 82.9

5. Find CP when SP = Rs. 40.60, gain = 16%?

(a) Rs. 35 (b) Rs. 50

(c) Rs. 75 (d) Rs. 89

6. Find CP when SP = Rs. 51.70, loss = 12%?
(a) Rs. 58.75 (b) Rs. 62.25
(c) Rs. 65 (d) Rs. 69.27

7. A person incurs 5% loss by selling a watch for Rs. 1140. At what price should the watch be sold to earn 5% profit?
(a) Rs. 1380 (b) Rs. 1160
(c) Rs. 1260 (d) Rs. 1400

8. If the cost price is 96% of the selling price, then what is the profit percent?
(a) 5.72% (b) 3.72%
(c) 8.92% (d) 4.17%

9. A dishonest dealer professes to sell his goods at cost price but uses a weight of 960 gms instead of a Kg weight. Find his gain%?
(a) $\frac{27}{4}\%$ (b) $\frac{8}{3}\%$
(c) $\frac{25}{6}\%$ (d) $\frac{21}{4}\%$

10. A man sold two cows at Rs. 1995 each. On one he lost 10% and on the other he gained 10%. What his gain or loss percent?
(a) 4% (b) 2%
(c) 0.5% (d) 1%

11. Two discounts of 40% and 20% equal to a single discount of?
(a) 48% (b) 53%
(c) 52% (d) 60%

12. Amit buys 5 watches for Rs. 9450 and later sells them for Rs. 9700. How much profit does Amit make per watch?
(a) Rs. 75 (b) Rs. 80
(c) Rs. 60 (d) Rs. 50

13. The price of 12 chair and 8 table is Rs. 676. What is the price of 21 chair and 14 table?
(a) Rs. 1183 (b) Rs. 4732
(c) Rs. 1180 (d) Cannot be determine

14. Aditya sold TV to Sanjay at 12% more than the CP. If Sanjay paid Rs. 17696 for that TV then what was the original price of the TV?
(a) Rs. 15,500 (b) Rs. 15,820
(c) Rs. 15,520 (d) Rs. 15,800

15. Amit purchased 13 chair of Rs. 115 each and sold all at Rs. 1220. Then find the profit or Loss on the transaction
(a) Rs. 280 Loss (b) Rs. 275 Loss
(c) Rs. 325 Profit (d) Rs. 350 Profit

16. Aditya purchase a book with a 20% discount on the marked price. How much did he pay if the book marked was Rs. 500?
(a) Rs. 400 (b) Rs. 300
(c) Rs. 200 (d) Rs. 500

17. By selling a book for Rs. 360, 20% profit was earned. What is the CP of the book?
(a) Rs. 300 (b) Rs. 200
(c) Rs. 250 (d) Cannot be determined

18. Profit earned by selling an article at Rs. 1630 is same as the loss incurred by selling the article for Rs. 1320. What is the CP?
(a) Rs. 1475 (b) Rs. 1300
(c) Rs. 1350 (d) Rs. 1275

19. If the CP of 50 items is equal SP of 40 items then what is the profit or loss%?
(a) 20% (b) 15%
(c) 25% (d) 35%

20. If a banana's cost is Rs. 1.25 and apple's cost is Rs. 1.75 what will be the cost of 2 Dozen of Banana and 3 Dozen of apple?
(a) Rs. 93 (b) Rs. 83
(c) Rs. 85 (d) Rs. 70

21. Nutan bought a watch with 24% discount. If she pays Rs. 779 for that watch then what is the marked price of watch?
(a) Rs. 950 (b) Rs. 975
(c) Rs. 1000 (d) Rs. 1025

22. Nutan pays Rs. 2140 for 3 calculator and 4 Pen while he pay Rs. 1355 for an additional calculator and 5 Pen. Then what he paid for Calculator only?
(a) Rs. 175 (b) Rs. 480
(c) Rs. 655 (d) Can't be determined

23. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?
(a) 3 (b) 4
(c) 5 (d) 6

24. A man sold his two horses for Rs. 770 each, on one he gained 10% & on the other he lost 10%. The average gain or loss percentage is
(a) 100% (b) 0.96%
(c) 4% (d) 1%

25. If the selling price of an article is $\frac{4}{3}$rd of its cost price the profit in the transaction is
(a) $16\frac{2}{3}\%$ (b) $20\frac{1}{2}\%$
(c) $25\frac{1}{2}\%$ (d) $33\frac{1}{3}\%$

26. A sells his house worth Rs. 10 lakh to B at a loss of 10%. Later B sells it back to A at 10% profit. The result of the two transactions is

(a) A neither loses nor gains
(b) A loses Rs. 10,000
(c) A loses Rs. 2,00,000
(d) B gains Rs. 1,10,000

27. A fair price shopkeeper takes 10% profit on his goods. He lost 20% goods during theft his loss% is
(a) 8% (b) 10%
(c) 11% (d) 12%

28. Aditya bought 200 dozen orange at Rs. 10 per dozen and he spent Rs. 500 on transportation. He sold each orange at Rs. 1 each. What was his profit or loss%?
(a) 5% (b) 6%
(c) 4% (d) Can't be determine

29. If 11 Mango are bought for Rs. 10 and sold at 10 for Rs. 11. What was Gain or Loss%?
(a) 24% (b) 21%
(c) 26% (d) 25%

30. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is:
(a) Rs. 15 (b) Rs. 16
(c) Rs. 18 (d) Rs. 25

31. If selling price is doubled, the profit triples. Find the profit percent:
(a) $66\frac{2}{3}\%$ (b) 100%
(c) $105\frac{1}{3}\%$ (d) 120%

32. Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is:
(a) 30% (b) $33\frac{1}{3}\%$
(c) 35% (d) 44%

33. The cost price of 12 tables is equal to the selling price of 16 tables. The loss percent is
(a) 15% (b) 20%
(c) 25% (d) 30%

34. Two continuous discounts of 4% on any thing should be equal to
(a) 8% (b) 7.92%
(c) 7.84% (d) 8.08%

35. A sells a bicycle to B at a profit of 20% and B sells it to C at a profit of 25%. If C pays Rs. 1500, what did A pay for it?
(a) Rs. 825 (b) Rs. 1000
(c) Rs. 1100 (d) Rs. 1125

36. Ram purchases a chair at Rs. 70 and spent Rs. 17 on its repair and 50 paise on cartage. If he sold the chair at Rs. 100 then his approximate margin of profit will be?
(a) 13.30% (b) 11.25%
(c) 12.5% (d) 14.3%

37. A shopkeeper marks his goods 20% above CP but allows 30% discount for cash. His net loss is?
(a) 8% (b) 20%
(c) 10% (d) 16%

38. A single discount, equivalent to a successive discount of 40% and 30% is?
(a) 55% (b) 56%
(c) 57% (d) 58%

39. If the CP of 13 bats is Rs. 390. What is the price when it is sold at 10% loss?
(a) Rs. 200 (b) Rs. 300
(c) Rs. 350 (d) Rs. 351

40. If an item is sold for Rs. 924 then there is a profit of 10%. What is the cost price?
(a) Rs. 840 (b) Rs. 860
(c) Rs. 880 (d) Rs. 900

Moderate

1. The cost of an article including the sales tax is Rs. 616. The rate of sales tax is 10%, if the shopkeeper has made a profit of 12%, then the cost price of the article is?
(a) Rs. 350 (b) Rs. 500
(c) Rs. 650 (d) Rs. 800

2. Two third of consignment was sold at a profit of 5% and the remainder at a loss of 2%. If the total profit was Rs. 400, the value of the consignment was?
(a) Rs. 12000 (b) Rs. 14000
(c) Rs. 15000 (d) Cannot be determined

3. A tradesman gives 4% discount on the marked price and gives one article free for buying every 15 articles and thus gains 35%. The marked price is appox. how much percent above the CP?
(a) 20% (b) 30%
(c) 40% (d) 50%

4. When a producer allows 36% concession on the retail price of his product, he earns a profit of 8.8%. What would be his profit percent if the commission is reduced by 24%?
(a) 48.2% (b) 49%
(c) 49.6% (d) 51%

5. A person earns 15% on investment but loses 10% on another investment. If the ratio of the two investments be 3 : 5, what is the gain or loss on the two investments taken together?
(a) 0.625% (b) 0.8%
(c) 0.9% (d) 1.2%

6. The profit earned by selling an article for Rs. 900 is double the loss incurred when the same article is sold for Rs. 450. At what price should the article be sold to make 25% profit?
(a) Rs. 400 (b) Rs. 500
(c) Rs. 700 (d) Rs. 750

7. A shopkeeper sold some article at the rate of Rs. 35 per article and earned profit of 40%. At what price each article should have been sold so that profit of 60% was earned.
(a) Rs. 45 (b) Rs. 42
(c) Rs. 39 (d) Rs. 40

8. Due to a 20% rise in price of sugar, a bachelor is able to buy 1.5 kg less for Rs. 135. What is the increased price of sugar per kg?
(a) Rs. 15 (b) Rs. 21
(c) Rs. 18 (d) Rs. 24

9. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is:
(a) No profit, no loss (b) 5%
(c) 8% (d) 10%

10. A TV set is being sold for Rs. x in Chandigarh. A dealer went to Delhi and bought the TV at 20% discount (from the price of Chandigarh). He spent Rs. 600 on transport. Thus he sold the set in Chandigarh for Rs. x making $14\frac{2}{7}$% profit. What was x?
(a) Rs. 9600 (b) Rs. 8800
(c) Rs. 8000 (d) Rs. 7200

11. Sanjay purchased a chair marked at Rs. 800 at 2 successive discount of 10% and 15% respectively. He spent Rs. 28 on transportation and sold the chair for Rs. 800. How much is his gain percentage?
(a) 14% (b) 30%
(c) 25% (d) 40%

12. When a book is sold at its Marked Price it gives a profit of 40%. What will happen if it is sold at half the marked Price?
(a) 30% profit (b) 25% loss
(c) 30% loss (d) 40% profit

13. Aditya purchased 14 shirt & 25 pants at Rs. 45 and Rs. 55 each respectively. What should be the approximate overall average selling price of shirt and pant so that 40% profit is earned?
(a) Rs. 72.5 (b) Rs. 71
(c) Rs. 72 (d) Rs. 70

14. A person sells 36 apple per rupee and suffers a loss of 4%. Find how many apple per rupees to be sold to have a gain of 8%.
(a) 32 (b) 16
(c) 4 (d) 15

15. Ram sells an article to Girish at a gain 20%, Girish sells it to Sanjay at a gain of 10% and Sanjay sells it to Aditya at a gain of $12\frac{1}{2}$% . If aditya pay Rs. 59.40. What did it cost Ram?
(a) Rs. 40 (b) Rs. 22
(c) Rs. 24 (d) Rs. 18

16. A fruit seller sells $\frac{3}{5}$th part of fruit at a profit of 10% and remaining at a loss of 5%. If the total profit is Rs. 1500 then what is the total cost price of fruit?
(a) Rs. 37500 (b) Rs. 37000
(c) Rs. 36500 (d) Rs. 36000

17. A shopkeeper buys a toy at Rs. 100 and sells it at Rs. 125. Another shopkeeper buys the same toy at Rs. 125 but sells it at Rs. 100. What are the respective profit and loss percentages for the two shopkeepers.
(a) 20%, 20% (b) 25%, 20%
(c) 25%, 25% (d) 25%, $16\frac{2}{3}$%

18. The marked price of a bed is Rs. 2400. The shopkeeper gives successive discounts of 10% and X% to customer. If the customer pays Rs. 1836 for the bed then find the value of X?
(a) 15% (b) 18%
(c) 12% (d) 10%

19. Three successive discounts of 10%, 12% and 15% will amount to a single discount of?
(a) 36.28% (b) 32.68%
(c) 34.68% (d) 37%

20. A loss of 19% gets converted into profit of 17% when the selling price is increased by Rs. 162. Find the cost price of the article.
(a) Rs. 300 (b) Rs. 350
(c) Rs. 400 (d) Rs. 450

21. A shopkeeper sold an article offering discount of 5% and earn a profit of 23.5%. What would have been the percentage of profit earned if no discount has been offered?
(a) 23% (b) 30%
(c) 33% (d) Cannot be determined

22. If Aditya sells an article to Nutan at 10% gain, while Nutan sells it to Manish at 20% gain at Rs. 1914 then what is the Cost Price?
(a) Rs. 1450 (b) Rs. 1340
(c) Rs. 1560 (d) Rs. 1780

23. Rita buys an article for Rs. 9600. She sells it at 12% loss and gets some money and from that money she again buys an article and this time she gets 12% profit. What is the profit or loss she gets from this transaction?
(a) Rs. 130 (b) Rs. 138
(c) Rs. 138.24 (d) Rs. 138.42

24. Nutan bought 30 dozens of oranges for her juice stall in the school fair. She paid Rs. 8 per dozen of oranges. She also had to pay Rs. 500 as the stall fee to the school authorities. She calculated that each glass of juice would need 3 oranges. How much should she charge per glass of juice so as to make 20% profit?
(a) Rs. 7.20 (b) Rs. 7.40
(c) Rs. 7.60 (d) Rs. 7.80

25. Aditya bought a scooter for a certain sum of Money. He spend 15% of cost price on repair and sold it for a profit on Rs. 1104. What is the CP of Scooter?
(a) Rs. 600 (b) Rs. 700
(c) Rs. 800 (d) Rs. 900

26. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?
(a) 30% (b) 70%
(c) 100% (d) 250%

27. A shopkeeper sells one transistor for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. His total gain or loss percent is:
(a) $5\frac{15}{17}$% loss (b) $5\frac{15}{17}$% gain
(c) $6\frac{2}{3}$% gain (d) $6\frac{1}{2}$% Loss

28. A dealer buys a table listed at Rs. 600 and gets successive discount of 10% and 20%. What is his profit or loss percent if he sells it at Rs. 540?
(a) 25% (b) 20%
(c) 15% (d) $17\frac{1}{2}$%

29. Sonal buys mangoes at the rate of 3 kgs for Rs. 21 and sells them at 5 kgs for Rs. 50. To earn a profit of Rs. 102, he must sell how many mangoes?
(a) 34 kgs (b) 52 kgs
(c) 26 kgs (d) 32 kgs

30. An electric pump was sold at a profit of 15%. Had it been sold for Rs. 600, the profit would have been 20%. The former selling price is
(a) Rs. 500 (b) Rs. 540
(c) Rs. 575 (d) Rs. 600

31. On a Rs. 10,000 payment order, a person has a choice between three successive discounts of 10%, 10% and 30% and three successive discounts of 40%, 5% and 5%. By choosing the better offer, he can save?
(a) Rs. 200 (b) Rs. 225
(c) Rs. 400 (d) Rs. 255

32. A man sells each of his 2 articles for Rs. 99. On one he gains 10% and on the other he incurred a loss of 1%. What is his total gain?
(a) 9% (b) $4\frac{4}{19}$%
(c) 4.5% (d) 5.5%

33. The market price of an article is Rs. 100. If the article is sold at a discount of 10%, then 35% profit is realised. How much loss or profit will be made if it is sold for Rs. 30 less than market price?
(a) 5% loss (b) 8% gain
(c) 5% gain (d) 8% loss

34. A shopkeeper sold an article offering discount of 24% and earn a profit of 23.5%. What would have been the percentage of profit earned if no discount had been offered?
(a) 63% (b) 62.50%
(c) 60% (d) Cannot be determined

35. On selling an article for Rs. 500 the loss accrued is 20%. To make a profit of 20% the article must be sold at?
(a) Rs. 700 (b) Rs. 750
(c) Rs. 800 (d) Rs. 900

36. The CP of 19 article is equal to the selling price of 15 article. Gain % is?
(a) 26% (b) $26\frac{1}{3}$%
(c) 12% (d) $26\frac{2}{3}$%

37. The cash difference between the selling price of an article at a profit of 4% and 6% is Rs. 3. The ratio of the 2 selling price is?
(a) 50 : 53 (b) 52 : 53
(c) 51 : 53 (d) Cannot be determined

38. A shopkeeper has to sell 24 Kg of sugar. He sells a part of these at a gain of 20% and the rest at a loss of 5%. If on the whole he earns a profit of 10%, the amount of Sugar sold at a loss is?
(a) Rs. 7.5 (b) Rs. 9.6
(c) Rs. 10 (d) Cannot be determined

39. If a shopkeeper sells 25 articles at Rs. 45 per article after giving 10% discount and earns 50% profit. If the discount is not given the profit gained is?

(a) 66.67% (b) 32%
(c) 35% (d) Cannot be determined

40. The price of a TV is Rs. 10,000. If successive discount of 15%, 10% and 5% allowed. Then at what price does a customer buy?

(a) Rs. 7267.50 (b) Rs. 7000
(c) Rs. 7200 (d) Cannot be determined

Difficult

1. Sarita sells a Phone at a profit of 20%. If she had bought it at 20% less and sold it for Rs. 180 less, she would have gained 25%. Find the cost price of the Phone.

(a) Rs. 800 (b) Rs. 850
(c) Rs. 900 (d) Rs. 1000

2. Ravi purchases 90 pens and sells 40 pens at a gain of 10% and 50 pens at a gain of 20%. Had he sold all of them at a uniform profit of 15% he would have got Rs. 40 less. Find the cost price of each pen.

(a) Rs. 80 (b) Rs. 75
(c) Rs. 90 (d) Rs. 100

3. Savita buys 5 shirts and 10 pants for Rs. 1600. She sells shirts at a profit of 15% and pants at a loss of 10%. If her over all profit was Rs. 90, what was the cost price of a shirt and a pant?

(a) Rs. 175, Rs. 50 (b) Rs. 200, Rs. 50
(c) Rs. 200, Rs. 60 (d) Cannot be determined

4. At a cost of 60 paise per article, Sarika produces 750 articles. She puts the selling price such that if only 600 articles are sold, she would have made a profit of 40% on the outlay. However, 120 articles got spoilt and she was able to sell 630 articles at this price. Find her actual profit or loss percent as the percentage of total outlay assuming that the unsold articles are useless.

(a) 47% profit (b) 51% profit
(c) 36% loss (d) 28% loss

5. Kritika bought 25 i-pads and i-phones for Rs. 205000. She sold 80% of the i-pads and 12 i-phones for a profit of Rs. 40000. Each i-pad was marked up by 20% over cost and each i-phone was sold at a profit of Rs. 2000. The remaining i-pads and 3 i-phones could not be sold. What is Kritika's overall profit/loss?

(a) Rs. 500 profit (b) Rs. 1000 loss
(c) Rs. 1500 profit (d) no profit, no loss

6. Sasha goes to a shop to buy a sofa set and a center table. She bargins for a 10% discount on the center table and 25% discount on sofa set. However, the manager, by mistake, interchanged the discount percentage figures while making the bill and Sasha paid accordingly. When compared to what she should pay for her purchases, what percentage did Sasha pay extra given that the center table costs 40% as much as the sofa set.

(a) 7.1% (b) 7.5%
(c) 7.9% (d) 8.1%

7. Paras Health Care made 3000 strips of vitamin tablets at a cost of Rs. 4800. The company gave away 1000 strips of tablets to doctors as free samples. A discount of 25% is given on the printed price. Find the ratio of profit if the price is raised from Rs. 3.25 to Rs. 4.25 per strip and if at the latter price, samples to doctors were done away with.

(a) Rs. 36.7 (b) Rs. 49.3
(c) Rs. 63.5 (d) Rs. 71.7

8. APD printed 3000 copies of 'Career Power' at a cost of Rs. 240000. It gave 500 copies free to different philanthropic institutions. It allowed a discount of 25% on the published price and gave one copy free for every 25 copies bought at a time. It was able to sell all the copies in this manner. If the published price is Rs. 325, then what is its overall gain or loss percentage in the whole transaction?

(a) 89% gain (b) 120% loss
(c) 140% loss (d) 143.75% gain

9. Surbhi bought a combined total of 25 i-pads and i-phones. She marked up the i-pad by 20% on the cost price, while each i-phone was marked up by Rs. 2000. She was able to sell 75% of the i-pads and 2 i-phones and make a profit of Rs. 49000. The remaining i-pads and 3 i-phones could not be sold by her. Find her overall profit or loss if she gets no return on unsold items and it is known that an i-phone costs 50% of an i-pad.

(a) Gain of Rs. 48500 (b) Loss of Rs. 48500
(c) Gain of Rs. 51400 (d) no profit, no loss

10. A merchant buys 4000 kg of wheat, one-fifth of which he sells at a gain of 5 per cent, one-fourth at a gain of 10%, one-half at a gain of 12 percent, and the remainder at a gain of 16 percent. If he had sold the whole at a gain of 11 percent, he would have made Rs. 72.80 more. What was the cost price of the crop per kg?

(a) Rs. 2 (b) Rs. 2.60
(c) Rs. 2.50 (d) Rs. 2.80

11. Ajit calculates his profit percentage on the selling price whereas Rohit calculates his on the cost price. They find that the difference of their profits is Rs. 100. If the selling price of both of them are the same and both of them get 25% profit, find their selling price?

(a) Rs. 1200
(b) Rs. 1500
(c) Rs. 1800
(d) Rs. 2000

12. A pen was sold for a certain sum and there was a loss of 20%. Had it been sold for Rs. 12 more, there would have been a gain of 30%. What would be the profit if the pen was sold for Rs. 4.80 more than what it was sold for?
(a) 15%
(b) 23%
(c) 29%
(d) no profit, no loss

13. A white goods dealer pays 10% custom duty on an i-phone that costs Rs. 25000 in UK. For how much should he mark it, if he desires to make a profit of 20% after giving a discount of 25% to the buyer?
(a) Rs. 32000
(b) Rs. 38000
(c) Rs. 44000
(d) Cannot be determined

14. A cab driver makes a profit of 20% on every trip when he carries 3 passengers and the price of petrol is Rs.30 a litre. Find the percentage profit for the same journey if he goes with four passengers per trip and the price of petrol reduces to Rs. 24 a litre?
(Assume that revenue per passenger is the same in both the cases.)
(a) 100%
(b) 76%
(c) 54%
(d) 43%

15. Anil bought an item with $12\frac{1}{2}\%$ discount on the labelled price. He sold the item with $17\frac{1}{2}\%$ profit on the labelled price. What was his percent profit on the price he bought?
(a) 35%
(b) $34\frac{1}{7}\%$
(c) $34\frac{2}{7}\%$
(d) $35\frac{2}{7}\%$

16. Divyam bought an article with 15% discount on the labelled price. He sold the article with 10% profit on the labelled price. What was his percent profit on the price he bought?
(a) $28\frac{7}{17}\%$
(b) $29\frac{7}{17}\%$
(c) $29\frac{5}{17}\%$
(d) Data inadequate

17. A shopkeeper sold Chairs at Rs. 2139 each after giving 7% discount on labelled price. Had he not given the discount, he would have earned a profit of 15% on the cost price. What was the cost price of each chairs?
(a) Rs. 2500
(b) Rs. 2100
(c) Rs. 2000
(d) Rs. 1900

18. A shopkeeper sold decks at Rs. 166 each after giving 17% discount on labelled price. Had he not given the discount, he would have earned a profit of 25% on the cost price. What was the cost price of each deck?
(a) Rs. 165
(b) Rs. 155
(c) Rs. 160
(d) Rs. 164

19. A garment company declared 15% discount for wholesale buyers. Mr. Ashish bought garments from the company for Rs. 25000 after getting discount. He fixed up the selling price of garments in such a way that he earned a profit of 8% on original company price. What is the approximate total selling price?
(a) Rs. 28000
(b) Rs. 29000
(c) Rs. 31700
(d) Rs. 28500

20. A garment company declared 14% discount for wholesale buyers. Mr. Swami bought garments from the company for Rs. 860 after getting discount. He fixed up the selling price of garments in such a way that he earned a profit of 6% on original company price. What is the approximate total selling price?
(a) Rs. 1060
(b) Rs. 1160
(c) Rs. 960
(d) Cannot be determined

Previous Year (Memory Based)

1. A man buys 10 articles for Rs. 8 and sells them at the rate of Rs. 1.25 per article. His gain percent is:
(a) 50
(b) $56\frac{1}{4}\%$
(c) $19\frac{1}{2}\%$
(d) 20%

2. The profit earned by selling an article for Rs. 515 is equal to the loss incurred when the same article is sold for Rs. 475. What should be the selling price of the article for making 40% per cent profit?
(a) Rs. 693
(b) Rs. 707
(c) Rs. 683
(d) Rs. 673

3. A reduction of 10 per cent in the price of potatoes enables me to obtain 25 kg more for Rs. 225. What is the reduced price per kg? Find also the original price per kg.
(a) Rs 0.9, Re 1
(b) Re 1, Rs 2
(c) Rs. 0.8, Rs. 2
(d) None of these

4. A 25% hike in the price of tea forces a person to purchase 2 kg less for Rs. 75. Find the original price of the tea.

(a) Rs. 7 (b) Rs. 8
(c) Rs. 7.5 (d) Rs. 8.5

5. A person buys 12 eggs for Rs. 15 and sells them at 10 for Rs. 14. Wht does he gain or lose per cent?
(a) 10% loss (b) 12% profit
(c) 12% loss (d) 15% profit

6. A dishonest dealer sells the goods at 44% loss on cost price but uses 30% less weight. What is his percentage profit or loss?
(a) 20% gain (b) 28% gain
(c) 20% loss (d) 25% loss

7. A seller uses 1250 gm in place of one kg to sell his goods. Find his actual % profit or loss, when he sells his article on 25% gain on cost price?
(a) 25% loss (b) 15% gain
(c) Neither loss nor gain
(d) Can't be determined

8. A person sold a chair at a profit of 14%. Had he sold it for Rs. 15 more, 17% would have been gained. Find the cost price.
(a) Rs. 550 (b) Rs. 650
(c) Rs. 600 (d) Rs. 500

9. I sold a book at a loss of 8%. Had I sold it for Rs. 120 more, 7% would have been gained. Find the cost price.
(a) Rs. 900 (b) Rs. 800
(c) Rs. 600 (d) Rs. 750

10. An article is sold at a profit of 15%. If both the cost price and selling price are Rs. 56 less, the profit would be 7% more. Find the cost price.
(a) Rs. 176 (b) Rs. 186
(c) Rs. 156 (d) Rs. 196

11. An article is sold at 30% profit. If its cost price and selling price are less by Rs. 25 and Rs. 19 respectively the percentage profit increases by 20%. Find the cost price.
(a) Rs. 97 (b) Rs. 92.5
(c) Rs. 98.5 (d) Rs. 98

12. An article is sold at 45% profit. If its cost price is increased by Rs. 80 and at the same time if its selling price is also increased by Rs. 70, the percentage of profit becomes 30%. Find the cost price:
(a) Rs. 246 (b) Rs. $226\frac{2}{3}$
(c) Rs. $293\frac{1}{3}$ (d) Rs. $226\frac{1}{3}$

13. $\frac{1}{5}$ of a commodity is sold at 25% profit, $\frac{1}{10}$ is sold at 40% profit and the rest at 30% profit. If a profit of Rs. 150 is earned, then find the value of the commodity:
(a) Rs. 550 (b) Rs. 450
(c) Rs. 650 (d) Rs. 500

14. A person bought two horses for Rs. 960. He sold one at a loss of 20% and the other at a gain of 60% and he found that each horse was sold at the same price. Find the cost price of two horses:
(a) Rs. 640, Rs. 320 (b) Rs. 540, Rs. 420
(c) Rs. 440, Rs. 520 (d) Rs. 650, Rs. 310

15. $\frac{5}{8}$ of a consignment was sold at 16% loss and the rest at a profit of 8%. If there was an overall loss of Rs. 140, find the value of the consignment?
(a) Rs. 2000 (b) Rs. 2500
(c) Rs. 1800 (d) Rs. 2400

16. If goods be purchased for Rs. 380, and $\frac{2}{3}$ rd be sold at a loss of 15 per cent, at what gain per cent should the remainder be sold so as to gain 10 per cent on the whole transaction.
(a) 60% (b) 70%
(c) 90% (d) 40%

17. A man buys coffee for Rs. 7200. He sells $\frac{3}{8}$th of it at a profit of 24%. At what per cent gain should he sell remaining $\frac{5}{8}$th so as to make an overall profit of 21% on the whole transaction?
(a) $18\frac{1}{5}$% (b) $20\frac{1}{5}$%
(c) $19\frac{1}{5}$% (d) Data inadequate

18. Oranges are bought at 12 for a rupee and an equal number more at 8 for a rupee. If these are sold at 15 for a rupee, find the loss or gain per cent.
(a) 36% loss (b) 36% profit
(c) 38% loss (d) 38% profit

19. A person bought two tables for Rs. 860. He sold one at a gain of 5% and the other at a gain of 10% and he found that each table was sold at the same price. Find the cost prices of the two tables.
(a) Rs. 440, Rs. 420 (b) Rs. 460, Rs. 400
(c) Rs. 480, Rs. 380 (d) Data inadequate

20. Raman buys 5 pens and 30 pencils for Rs. 1000. He sells the pens at a profit of 15% and pencils at a profit of 10% and makes a total profit of Rs. 120. Find the cost of a pen and of a pencil.
(a) Rs. 70, Rs. 30 (b) Rs. 85, Rs. 15
(c) Rs. 80, Rs. 20 (d) Data inadequate

21. A person bought some oranges at the rate of 12 per rupee. He bought the same number of oranges at the rate of 8 per rupee. He mixes both the types and sells at 20 for rupees 2. In this business he bears a loss of Rs. 8. Find out how many oranges he bought in all?

(a) 1680 (b) 1820
(c) 1920 (d) 1290

22. A man buys two horses for Rs. 1550. He sells one so as to lose 23% and the other so as to gain 27%. On the whole he neither gains nor loses. What does each horse cost?

(a) Rs. 807, Rs. 743 (b) Rs. 817, Rs. 733
(c) Rs. 827, Rs. 723 (d) Rs. 837, Rs. 713

23. A trader bought a car at 5% discount on its original price. He sold it at 10% increase on the price he bought it. What percentage of profit did he make on the original price?

(a) 5.5% (b) 6.5%
(c) 4.5% (d) 3.5%

24. 20 kg of potato costs as much as 5 kg of tomato, 12 kg of tomato costs as much as 30 kg of onion, 15 kg of onion costs as much as 18 kg of cabbage. If 10 kg of cabbage costs Rs. 50. find the cost of 24 kg of potato.

(a) Rs. 90 (b) Rs. 72
(c) Rs. 108 (d) Rs. 96

25. A man sells two articles, each for the same price Rs. 640. He earns 20% profit on the first and 10% profit on the second. Find his overall per cent profit.

(a) 14.78% (b) 14.08%
(c) 14.58% (d) None of these

26. Miheer calculates his profit percentage on the selling price whereas Satya calculates his profit on the cost price. They find that the diffference of their profits is Rs. 110. If the selling price of both of them are the same, and Miheer gets 10% profit and Satya gets 5% profit, then find their selling price.

(a) Rs. 2100 (b) Rs. 2250
(c) Rs. 1750 (d) Rs. 2200

27. A man sells two horses for Rs. 11900. The cost price of the first is equal to the selling price of the second. If the first is sold at 30% loss and the second at 25% gain, what is his total gain or loss (in rupees)?

(a) Rs. 600 loss (b) Rs. 700 loss
(c) Rs. 750 gain (d) Rs. 700 gain

28. A dealer sells a table for Rs. 405, making a profit of 35%. He sells another table at a loss of 30%, and on the whole he makes neither profit nor loss. What did the second table cost him?

(a) Rs. 360 (b) Rs. 350
(c) Rs. 340 (d) Rs. 300

29. What will be the percentage profit on selling an article at a certain price if there is a loss of 25% when the article is sold at half of the previous selling price?

(a) 45% (b) 60%
(c) 55% (d) 50%

30. Jeevan bought an article with 30 per cent discount on the labelled price. He sold the article with 12 per cent profit on the labelled price. What was his per cent profit on the price he bought?

(a) 40 (b) 50
(c) 60 (d) Data inadequate

31. A shopkeeper sold an article for Rs. 210 after giving 16% discount on the labelled price and made 5% profit on the cost price. What would have been the percentage profit, had he not given the discount?

(a) 50% (b) 25%
(c) 30% (e) 20%

32. Alok bought 25 kg of rice of Rs. 6.00 per kg and 35 kg of rice at the rate of Rs. 7.00 per kg. He mixed the two and sold the mixture at the rate of Rs. 6.75 per kg. What was his gain or loss in this transaction?

(a) Rs. 16.00 gain (b) Rs. 16.00 loss
(c) Rs. 20.00 gain (d) Rs. None of these

33. Prateek sold a music system to Kartik at 20% gain and Kartik sold it to Swastik at 40% gain. If Swastik paid Rs. 10500 for the music system. What did Prateek pay for the music system?

(a) Rs. 8240 (b) Rs. 7500
(c) Rs. 6250 (d) Cannot be determined

34. The owner of a toy shop charges his customer 33% more than the cost price. If the customer paid Rs. 4921 for a toy, then what was the cost price of the toy?

(a) Rs. 3850 (b) Rs. 3700
(c) Rs. 3550 (d) Rs. 3900

35. Rani bought a piece of cloth for Rs. 950 and spent Rs. 300 on designing it. At what price should she sell it to make 30% profit?

(a) Rs. 1650 (b) Rs. 1550
(c) Rs. 1525 (d) Rs. 1625

36. Naresh purchased a TV set for Rs. 11250 after getting discount of 10% on the labelled price. He spent Rs. 150 on transport and Rs. 800 on installation. At what price should it be sold so that the profit earned would have been 15%, if no discount was offered?

(a) Rs. 12938 (b) Rs. 14030
(c) Rs. 13450 (d) Rs. 15467

37. The owner of an electronics shop charges his customer 22% more than the cost price. If a customer paid Rs. 10980 for a DVD player, then what was the cost price of the DVD player?

(a) Rs. 8000 (b) Rs. 8800
(c) Rs. 9500 (d) Rs. 9000

38. A company pays a rent of Rs. 25000 per month for office space to its owner. But if the company pays the annual rent at the beginning of the year, the owner gives a discount of 5% on the total annual rent? What is the annual amount the company pays to the owner after the discount?

(a) Rs. 285000 (b) Rs. 275000
(c) Rs. 300000 (d) Rs. 295000

39. Nandu purchased a mobile phone and got a disocunt of 5% from dealer. He sold that mobile phone at a profit of 10% on his purchase price. What is his percentage profit on actual price of the mobile phone?

(a) 2% (b) 3%
(c) 4.5% (d) 5%

40. In a sale, a perfume is available at a discount of 15% on the selling price. If the perfume's discounted selling price is Rs. 3675.4, what was the original selling price of the perfume?

(a) Rs. 4324 (b) Rs. 4386
(c) Rs. 4400 (d) Rs. 4294

Foundation

Solutions

1. **(a);** CP = Rs. 27.50, SP = Rs. 28.60

Then Gain = SP – CP = 28.60 – 27.50 = Rs. 1.10

Since, $\text{Gain\%} = \frac{\text{gain} \times 100}{\text{CP}}\%$

$\Rightarrow \text{Gain\%} = \frac{1.10 \times 100}{27.50} = 4\%$

2. **(b);** CP = Rs. 490, SP = Rs. 465.50

Loss = CP – SP = 490 – 465.50 = Rs. 24.50

$\text{Loss\%} = \frac{\text{loss} \times 100}{\text{CP}}\% = \frac{24.50 \times 100}{490} = 5\%$

3. **(b);** $\text{SP} = \left[\frac{100 + \text{gain\%}}{100}\right] \times \text{CP}$

$\Rightarrow \text{SP} = \left[\frac{100 + 20}{100}\right] 56.25 = \text{Rs. } 67.50$

4. **(c);** $\text{SP} = \left[\frac{100 - \text{loss\%}}{100}\right] \times \text{CP}$

$\Rightarrow \text{SP} = \left[\frac{100 - 5}{100}\right] \times 80.40 = \text{Rs. } 76.38$

5. **(a);** $\text{CP} = \frac{100 \times \text{SP}}{100 + \text{gain\%}}$

$\Rightarrow \text{CP} = \frac{100 \times 40.60}{100 + 16} = \text{Rs. } 35$

6. **(a);** $\text{CP} = \frac{100 \times \text{SP}}{100 - \text{loss\%}}$

$\Rightarrow \text{CP} = \frac{100 \times 51.70}{100 - 12} = \text{Rs. } 58.75$

7. **(c);** Let the new SP be Rs. x then

$\frac{100 - \text{loss\%}}{\text{1st SP}} = \frac{100 + \text{gain\%}}{\text{2nd SP}}$

(CP is same in both case)

$\frac{100 - 5}{1140} = \frac{100 + 5}{x}, \quad x = \frac{105 \times 1140}{95} = \text{Rs. } 1260$

8. **(d);** Let SP = Rs. 100 then CP = Rs. 96

Profit = SP – CP = 100 – 96 = Rs. 4

$\text{Profit\%} = \frac{\text{profit}}{\text{CP}} \times 100\% = \frac{4}{96} \times 100 = 4.17\%$

9. **(c);** Here, True weight = 1000g.

False weight = 960g.

Error change = (1000 – 960)g. = 40g.

$\Rightarrow \text{Gain\%} = \frac{\text{Error change}}{\text{True weight} - \text{Error}} \times 100\%$

$= \frac{40}{1000 - 40} \times 100\% = \frac{25}{6}\%$

10. **(d);** Here, since both gain and loss percent is same, hence the resultant value would be loss percent only.

$\Rightarrow \text{Loss\%} = \frac{a^2}{100} \quad [\text{where } a = 10\%]$

$= 1\%$

11. **(c);** Using net discount formula

$\Rightarrow \left[a + b - \frac{ab}{100}\right]\%$

Here, a = 40%, b = 20%

Applying both values in above formula:

$\Rightarrow \left[40 + 20 - \frac{40 \times 20}{100}\right]\% = 52\%$

12. (d); Using simple formula of

Profit = SP – CP = 9700 – 9450 = Rs. 250

[Total 5 watches]

$$\text{Profit on 1 watch} = \frac{\text{Rs. } 250}{5} = \text{Rs. } 50$$

13. (a); Here, cost of 12 chairs and 8 tables = Rs. 676

On dividing above equation by 4

$\Rightarrow$ Cost of 3 chairs and 2 tables = Rs. 676 × $\frac{1}{4}$

Now multiply it by 7

$\Rightarrow$ Cost of 21 chairs and 14 tables

= Rs. 676 × $\frac{7}{4}$ = Rs. 1183

14. (d); Let CP be Rs. 100

Then SP = Rs. 112. (12% more than CP)

$\Rightarrow$ Now if SP = Rs. 17696

Then by unitary method:

$$\Rightarrow \text{CP} = \frac{100}{112} \times 17696 = \text{Rs. } 15800$$

15. (b); Total SP given = Rs. 1220

Total CP of 13 chairs = Rs. 13 × 115 = Rs. 1495

$\Rightarrow$ Hence, CP > SP

$\Rightarrow$ Loss = CP – SP = Rs. 1495 – 1220 = Rs. 275

16. (a); Here, MP = Rs. 500

Now since we need discount of 20%

$$\Rightarrow \text{Amount paid} = \text{Rs.}\left[500 - 500 \times \frac{20}{100}\right] = \text{Rs. } 400$$

17. (a); Here, profit% = 20%

$$\Rightarrow \text{P\%} = \left[\frac{\text{SP} - \text{CP}}{\text{CP}}\right] \times 100\% \Rightarrow \frac{20}{100} = \frac{360 - \text{CP}}{\text{CP}}$$

$\Rightarrow$ CP = Rs. 300

18. (a); Here, profit = loss ...(i)

$\Rightarrow$ Here, profit = $(\text{SP})_1 - (\text{CP})$

and, Loss = $(\text{CP}) - (\text{SP})_2$

Now putting these values in (i)

$(\text{SP})_1 - (\text{CP}) = (\text{CP}) - (\text{SP})_2$

$$\text{CP} = \frac{(\text{SP})_1 + (\text{SP})_2}{2} = \text{Rs. } \frac{1630 + 1320}{2} = \text{Rs. } 1475$$

19. (c); As, CP of 50 items = SP of 40 items

$\Rightarrow$ 50 × (CP of 1 item) = 40 × (SP of 1 item)

$$\Rightarrow \frac{\text{CP of 1 item}}{\text{SP of 1 item}} = \frac{40}{50} = \frac{4}{5}$$

$$\text{Profit\%} = \frac{\text{SP} - \text{CP}}{\text{CP}} = \frac{5 - 4}{4} \times 100 = 25\%$$

20. (a); Cost of 1 banana = Rs. 1.25

Cos of 1 apple = Rs. 1.75

Cost of 2 dozen banana = Rs. 24 × 1.25 = Rs. 30

Cost of 3 dozen apple = Rs. 36 × 1.75 = Rs. 63

Total cost = Rs. (30 + 63) = Rs. 93

21. (d); The formula to determine MP of watch if we are given SP and discount% is:

$$\Rightarrow \left[\frac{\text{SP}}{100 - \text{D\%}} \times 100 = \text{MP}\right]$$

$$\text{MP} = \frac{779}{76} \times 100 = \text{Rs. } 1025$$

22. (b); Let cost of 1 calculator be Rs. 'C'

and cost of 1 pen be Rs. 'P'

According to question:

3C + 4P = 2140 ...(i)

1C + 5P = 1355 ...(ii)

Solving (i) and (ii)

We get C = Rs. 480 [cost of 1 calculator]

23. (c); CP of 6 toffees = Rs. 1, CP of 1 toffee = Rs. $\frac{1}{6}$

SP of x toffees = Rs. 1

(where x is no. of toffees to sell)

SP of 1 toffee = Rs. $\frac{1}{x}$

$$\text{Gain\%} = \frac{20}{100} = \frac{\frac{1}{x} - \frac{1}{6}}{\frac{1}{6}} \Rightarrow \frac{1}{5} \times \frac{1}{6} = \frac{1}{x} - \frac{1}{6} \Rightarrow x = 5$$

24. (d); Using Net Loss formula $= \frac{a^2}{100}\%$ [when P% = L%]

$$= \frac{(10)^2}{100}\% = 1\% \text{ Loss}$$

25. (d); Let CP = Rs. 1

$$\text{SP} = \text{Rs. } \frac{4}{3} \text{ (Given)}, \ \text{Profit} = \frac{4}{3} - 1 = \frac{1}{3}$$

$$\text{Profit\%} = \frac{1}{3} \times 100 = 33\frac{1}{3}\%$$

26. (b); Here, Net Loss % $= \left(\frac{a}{10}\right)^2 \% = 1\%$ Loss

$\Rightarrow$ So, Loss of 1% on Rs. 10 lakh = Rs. 10,000

27. (d); Using net effective formula,

$$\Rightarrow 10 - 20 - \frac{10 \times 20}{100} = 12\% \text{ Loss}$$

Hence, 12% Loss

28. (c); Total CP = 200 × 10 + 500 = Rs. 2500

Total SP = 1 × 200 × 12 = Rs. 2400

$$\%loss = \frac{100}{2500} \times 100 = 4\%$$

29. (b); CP of 11 Mangoes = Rs. 10

$$\Rightarrow \text{CP of 10 Mangoes} = \text{Rs.}\left[10 \times \frac{10}{11}\right] = \text{Rs.}\ \frac{100}{11}$$

SP of 10 Mangoes = Rs. 11

$$\%profit = \frac{11 - \frac{100}{11}}{\frac{100}{11}} \times 100\% = 21\%$$

30. (b); CP of 20 articles = SP of x articles

20 × CP of 1 article = x × SP of 1 article

$$\Rightarrow \frac{\text{CP of 1 article}}{\text{SP of 1 article}} = \frac{x}{20}$$

$$Profit\% = \frac{25}{100} = \frac{SP - CP}{CP} \Rightarrow \frac{1}{4} = \frac{20 - x}{x}$$

x = Rs. 16

31. (b); Let profit be P

Now, SP – CP = P ...(i)

In given question, when SP is doubled, P get tripled

2SP – CP = 3P ...(ii)

On solving (i) and (ii)

We get CP = P and SP = 2P

$$Profit\% = \frac{2P - P}{P} \times 100 = 100\%$$

32. (d); CP of 6 articles = Rs. 5, CP of 5 articles = Rs. $\frac{25}{6}$

SP of 5 articles = Rs. 6

$$\%gain = \frac{6 - \frac{25}{6}}{\frac{25}{6}} \times 100 = \frac{11}{25} \times 100 = 44\%$$

33. (c); CP of 12 tables = SP of 16 tables

$$\frac{\text{CP of 1 table}}{\text{SP of 1 table}} = \frac{16}{12} = \frac{4}{3}$$

$$\%Loss = \frac{4-3}{4} \times 100 = 25\%$$

34. (c); Using net discount formula $= \left[a + b - \frac{ab}{100}\right]\%$

where a = b = 4%

$$\Rightarrow \left[4 + 4 - \frac{4 \times 4}{100}\right]\% = 7.84\%$$

35. (b); Let CP for A be Rs. 100

A sells it to B at 20% profit

Rs. [100 + 20% of 100] = Rs. 120

Now B sells it to C at 25% profit

Rs. [120 + 25% of 120] = Rs. 150

If C buys at Rs. 150, A bought at Rs. 100

Hence, by unitary method,

If C bought at Rs. 1500, A paid = Rs. $\left[\frac{100}{150} \times 1500\right]$

= Rs. 1000

36. (d); Total CP = Rs. [70 + 17 + 0.5] = Rs. 87.50

SP = Rs. 100

$$Profit\% = \frac{12.50}{87.50} \times 100 = 14.3\%$$

37. (d); Let CP = Rs. 100, MP = Rs. 120

The SP after discount $= 120 \times \frac{70}{100} = 84$ Rs.

[30% discount]

So loss = 16% [CP – SP]

38. (d); Using net discount formula

$$\left[40 + 30 - \frac{40 \times 30}{100}\right] = 58\%$$

39. (d); Given Loss% = 10%

$$\Rightarrow 10\% = \left(\frac{CP - SP}{CP}\right) \times 100$$

$$\frac{10}{100} = \frac{390 - SP}{390},\ SP = \text{Rs. } 351$$

40. (a); Here, Profit% = 10%

$$\Rightarrow \frac{10}{100} = \frac{SP - CP}{CP} \Rightarrow \frac{1}{10} = \frac{924 - CP}{CP}$$

11CP = 9240 ⇒ CP = Rs. 840

Moderate

1. **(b);** 110% of SP = 616 (Rate of sales tax = 10%)

$$SP = \frac{616 \times 100}{110} = Rs.\ 560,\ CP = \frac{100 \times SP}{100 + gain\%}$$

$$CP = \frac{100 \times 560}{100 + 12} = Rs.\ 500$$

2. **(c);** Let the total value be Rs x

then value of $\frac{2}{3}$rd = Rs. $\frac{2x}{3}$,

value of $\frac{1}{3}$rd = Rs. $\frac{x}{3}$

According to question

$$\frac{2}{3}x\left(\frac{5}{100}\right) - \frac{1}{3}x\left(\frac{2}{100}\right) = 400$$

$$\frac{x}{30} - \frac{x}{150} = 400$$

$$\frac{5x - x}{150} = 400 \Rightarrow x = Rs.\ 15000$$

3. **(d);** Let the CP of each article be Rs. 100
Then CP of 16 articles = Rs. (100 × 16) = 1600

$$SP \text{ of } 16 \text{ articles} = 1600 \times \frac{135}{100} = Rs.\ 2160$$

(1 article free)

$$SP \text{ of each article} = \frac{2160}{15} = Rs.\ 144$$

If SP is Rs. 96, marked price = Rs. 100
If SP is Rs. 144, marked price

$$= \frac{100}{96} \times 144 = Rs.\ 150$$

Therefore marked price ≈ 50% above CP

4. **(c);** Let retail price = Rs. 100
then, Commission = Rs. 36
SP = retail price – concession = 100 – 36 = Rs. 64
But profit = 8.8%

$$CP = \frac{100 \times SP}{100 + gain\%} = \frac{100 \times 64}{100 + 8.8} = Rs.\ \frac{1000}{17}$$

New commission = Rs. 12 then
New SP = 100 – 12 = Rs. 88

$$Gain = 88 - \frac{1000}{17} = Rs.\ \frac{496}{17}$$

$$Gain\% = gain \times \frac{100}{CP} = \frac{\frac{496}{17}}{\frac{1000}{17}} \times 100 = 49.6\%$$

5. **(a);** Let the investments be Rs. 3x and Rs. 5x
Then total investment = 8x
Total receipt = 115% of 3x + 90% of 5x

$$= 115 \times \frac{3x}{100} + 90 \times \frac{5x}{100} = 7.95x$$

Loss = CP – SP = 8x – 7.95x = 0.05x

$$loss\% = 0.05x \times \frac{100}{8x} = 0.625\%$$

6. **(d);** Let CP be Rs. x
then, 900 – x = 2 (x – 450) [Profit = 2 Loss]
3x = 1800 ⇒ x = Rs. 600
CP = Rs. 600, gain required = 25%

$$SP = (100 + gain\%) \times \frac{CP}{100}$$

$$SP = (100 + 25) \times \frac{600}{100} = Rs.\ 750$$

7. **(d);** Here initially SP of some article = Rs. 35
Profit% = 40%
Now, finally SP of articles = Rs. x
Profit% = 60%
Here, CP is same in each case
⇒ $(CP)_1 = (CP)_2$

$$\Rightarrow \frac{(SP)_1}{100 + P_1\%} = \frac{(SP)_2}{100 + P_2\%},\ \frac{35}{140} = \frac{x}{160},\ x = Rs.\ 40$$

8. **(c);** Let price of sugar per kg is x so

$$\frac{135}{x} - \frac{135}{1.2x} = 1.5 \quad \text{[as given in question]}$$

$$135\left(\frac{0.2}{1.2x}\right) = 1.5,\ x = \frac{135 \times 0.2}{1.2 \times 1.5} = Rs.\ 15 \text{ per kg}$$

Increased price = 15 × 1.2 = Rs. 18 per kg

9. **(b);** Total CP of mixture = 26 × 20 + 30 × 36
520 + 1080 = Rs. 1600, SP = 30 × 56 = Rs. 1680

$$\%profit = \frac{80}{1600} \times 100 = 5\%$$

10. **(c);** The SP of TV in chandigarh = Rs. x
The dealer bought it at delhi at = Rs. 0.8x
[Discount of 20%]
Total CP of TV set (including transportation cost)
Rs. 0.8x + 600

$$\text{Given Profit\%} = 14\frac{2}{7}\%$$

$$\frac{100}{7}=\left(\frac{(x)-(0.8x+600)}{(0.8x+600)}\right)\times 100$$

$$\frac{1}{7}=\left(\frac{0.2x-600}{0.8x+600}\right),\ \text{On solving, } x = \text{Rs. } 8000$$

11. (c); MP of chair = Rs. 800
After getting successive discount of 10% and 15% respectively

$$\Rightarrow \text{CP of chair} = \text{Rs.}\left[800\times\frac{90}{100}\times\frac{85}{100}\right] = \text{Rs. } 612$$

Total CP (including transportation cost)
612 + 28 = Rs. 640

$$\text{Profit\%}=\frac{800-640}{640}\times 100 = 25\%$$

12. (c); Let us assume that cost of the book is Rs. 100 and Market Price is Rs. 140
If we sell the book at half of MP

$$\text{then selling Price}=\frac{140}{2}=\text{Rs. } 70$$

So percent loss = (100 – 70) = 30% loss

13. (c); Price of 14 shirt = 14 × 45 = Rs. 630
25 pant = 25 × 55 = Rs. 1375
Total price of 39 items = Rs. 2005

$$\text{Price}=\frac{2005}{39}\times 1.40 \qquad [\text{Overall profit} = 40\%]$$

$$= 71.97 = \text{Rs. } 72 \text{ (Approx)}$$

14. (a); Here, CP is same in both transactions
$(CP)_1 = (CP)_2$

$$\frac{(SP)_1}{100-x}=\frac{(SP)_2}{100+y} \qquad \left[\begin{aligned}\text{Where } x &= 4\%\text{ Loss}\\ y &= 8\%\text{ Gain}\end{aligned}\right]$$

$$(SP)_1 \text{ of 1 apple} = \text{Rs. }\frac{1}{36}$$

$$\Rightarrow \frac{\frac{1}{36}}{100-4}=\frac{(SP)_2}{100+8},\ (SP)_2=\frac{108}{96}\times\frac{1}{36}=\text{Rs. }\frac{1}{32}$$

Hence, in a rupee, the person can sell 32 apples.

15. (a); Here, Aditya paid Rs. 59.40 for article
But before that there were 3 transactions of gain 20%, 10% and 12.5%
So, initially Ram would have bought article at Rs. x

$$\Rightarrow x\times\frac{100+20}{100}\times\frac{100+10}{100}\times\frac{100+12.5}{100} = \text{Rs. } 59.40$$

$$x=\frac{59.40\times 100\times 100\times 100}{120\times 110\times 112.5} = \text{Rs. } 40$$

16. (a); Let CP of whole fruit = Rs. A
He sold $\frac{3}{5}$th part at 10% profit and remaining $\frac{2}{5}$th part at 5% loss
Total profit = Rs. 1500

$$1500=\left[\frac{3}{5}\times A\times\frac{10}{100}-\frac{2}{5}\times A\times\frac{5}{100}\right]$$

On solving above equation we get:
Total CP = A = Rs. 37,500

17. (b); **Case – 1**
Here, CP = Rs. 100, SP = Rs. 125
Profit of 1st shopkeeper = 25%
Case – 2
Here, CP = Rs. 125, SP = Rs. 100

$$\text{Loss of 2nd shopkeeper}=\left(\frac{125-100}{125}\right)\times 100\%$$

$$= 20\%$$

18. (a); Here, the successive discounts given are 10% and x%

$$2400\times\frac{90}{100}\times\frac{(100-x)}{100}=1836$$

$$100-x=\frac{1836\times 100\times 100}{2400\times 90}=85$$

So discount = 100 – 85 = 15%

19. (b); Three successful discount equivalent to

$$\left[x+y+z-\frac{xy+yz+zx}{100}+\frac{xyz}{(100)^2}\right]\%$$

$$(\text{where } x=10\%, y=12\%, z=15\%)$$

$$\left[10+12+15-\frac{10\times 12+12\times 15+15\times 10}{100}+\frac{10\times 12\times 15}{(100)^2}\right]\%$$

37 – 4.50 + 0.18 = 32.68%

20. (d); Let the original price be Rs. x
⇒ Loss of 19% = Rs. (x – 0.19x)
= Rs. 0.81x (old price)
⇒ Profit of 17% = Rs. (x + 0.17x)
= Rs. 1.17x (new price)
(New price) – (Old price) = Rs. 162
[According to given question]

$$1.17x - 0.81x = 162,\ x=\frac{162}{0.36}=\text{Rs. } 450$$

21. **(b);** Let CP = Rs. 100, Then SP = Rs. 123.5
Here discount of 5% is given
Let Mark Price be x Rs. then

$$x = \frac{123.50}{95} \times 100 = \text{Rs. } 130$$

So profit on Marked Price = 130 – 100 = 30%

22. **(a);** Let the original price bought by Aditya = Rs. 100

Aditya		Nutan		Manish
100	$\xrightarrow{10\%\text{ gain}}$	110	$\xrightarrow{20\%\text{ gain}}$	132

If Manish bought goods at Rs. 132
Initial CP = Rs. 100
Now, Manish bought goods at Rs. 1914

$$\text{Initial CP} = \text{Rs.}\left[\frac{100}{132} \times 1914\right] = \text{Rs. } 1450$$

23. **(c);** Initial CP = Rs. 9600

$$\text{SP after selling it at 12\% Loss} = \text{Rs. } 9600 \times \frac{88}{100}$$

New CP = Rs. 8448
Final SP after selling new CP price at 12% gain

$$\Rightarrow \text{Rs. } 8448 \times \frac{112}{100} = \text{Rs. } 9461.76$$

So, Total Loss = Initial CP – Final SP
= Rs. [9600 – 9461.76] = Rs. 138.24

24. **(b);** Total CP
= [CP of 30 dozen orange] + [CP of stall fee]
$\Rightarrow$ 8 × 30 + 500 = Rs. 740
Here, Nutan calculated that each glass needs 3 oranges and she wants to make 20% profit

$$\text{Per glass price} = \frac{740}{30 \times 12} \times 3 \times 1.20 = \text{Rs. } 7.40$$

25. **(c);** Let CP = Rs. 100
then total CP after repair = 100 + 15 = Rs. 115
after getting 20% profit
SP = 115 × 1.20 = 138 Rs.
but SP given is Rs. 1104

$$\text{so CP} = \frac{100}{138} \times 1104 = \text{Rs. } 800$$

26. **(b);** Let CP = Rs. 100
After 320% profit SP = Rs. 420
After increasing cost the,
CP = Rs. 125 [25% cost increase]
Profit = 420 – 125 = Rs. 295

$$\frac{295}{420} \times 100 = 70\% \text{ (Appx.)}$$

27. **(b);** $\text{CP of 1}^{\text{st}}\text{ item} = \frac{840}{1.2} = \text{Rs. } 700$ [gain of 20%]

$\text{CP of 2}^{\text{nd}}\text{ item} = \frac{960}{0.96} = \text{Rs. } 1000$ [loss of 4%]

Total CP = 700 + 1000 = Rs. 1700
SP = 840 + 960 = Rs. 1800

$$\%\text{profit} = \frac{100}{1700} \times 100 = 5\frac{15}{17}\%$$

28. **(a);** Given MP = Rs. 600
Hence on giving successive discounts of 10% and 20%,

$$\text{CP} = 600 \times \frac{90}{100} \times \frac{80}{100} = \text{Rs. } 432$$

$$\%\text{profit} = \frac{108}{432} \times 100 = 25\%$$

29. **(a);** $\text{CP of each kg mango} = \text{Rs. } \frac{21}{3} = \text{Rs. } 7$

$$\text{SP of each kg mango} = \text{Rs. } \frac{50}{5} = \text{Rs. } 10$$

Profit = SP – CP = Rs. 3
Here, Rs. 3 is profit earned for 1 kg
Similarly, Rs. 102 is profit earned for:

$$= \frac{1}{3} \times 102 = 34 \text{ kg}$$

30. **(c);** $\text{Old Profit\%} = \frac{15}{100} = \frac{(SP)_1 - CP}{CP}$ ...(i)

$$\text{New Profit\%} = \frac{20}{100} = \frac{(SP)_2 - CP}{CP} \quad \text{...(ii)}$$

[Here, $(SP)_2$ = Rs. 600]
= From (ii), we get CP = Rs. 500
Divide (i) and (ii):

$$\frac{3}{4} = \frac{(SP)_1 - 500}{600 - 500}$$

Hence, $(SP)_1$ = Former Selling price = Rs. 575

31. **(d);** Let us assume payment order be Rs. 100
Case – 1: Successive discount of 10%, 10%, 30%

$$\Rightarrow 100 \times \frac{90}{100} \times \frac{90}{100} \times \frac{70}{100} = \text{Rs. } 56.7$$

Case – 2: Successive discount of 40%, 5%, 5%

$$\Rightarrow 100 \times \frac{60}{100} \times \frac{95}{100} \times \frac{95}{100} = \text{Rs. } 54.15$$

For Rs. 100, person can save Rs. (56.7 – 54.15)
= Rs. 2.55

Hence, for Rs. 10000, he can save

$= \text{Rs. } \frac{2.55}{100} \times 10000 = \text{Rs. } 255$

32. (b); CP of 1st article $= \frac{99}{1.10} = \text{Rs. } 90$ [Profit of 10%]

CP of 2nd article $= \frac{99}{0.99} = \text{Rs. } 100$ [Loss of 1%]

CP of both article together = 100 + 90 = Rs. 190

SP of both article together = 99 + 99 = Rs. 198

$\%\text{profit} = \frac{198-190}{190} \times 100 = 4\frac{4}{19}\%$

33. (c); Here, (100 + Profit)% of CP

= Rs. (MP – 10% of MP)

(100 + 35)% CP = Rs. (100 – 10)

(135%) CP = Rs. 90 $\Rightarrow$ CP = Rs. $\frac{200}{3}$

SP of article (at Rs. 30 less than MP) = Rs. 70

$\text{Profit\%} = \frac{70 - \frac{200}{3}}{\frac{200}{3}} \times 100 = 5\%$

34. (b); Let CP = Rs. 100

Then SP = Rs. 123.5

Let Marked Price be x Rs., then $x = \frac{123.50}{76} \times 100$

[on discount of 24%]

= Rs. 162.50

So profit on Marked Price = 162.50 – 100 = 62.50%

35. (b); $\text{Loss} = \frac{20}{100} = \frac{CP - 500}{CP}$

CP = Rs. 625

Now Profit $= \frac{20}{100} = \frac{SP - 625}{625}$, SP = Rs. 750

36. (d); Here, CP of 19 article = SP of 15 article

$\frac{\text{CP of 1 article}}{\text{SP of 1 article}} = \frac{15}{19}$

$\%\text{gain} = \frac{19-15}{15} \times 100 = 26\frac{2}{3}\%$

37. (b); Here, we need to determine only ratio of selling price only.

SP is directly proportional to profit%

$= \frac{(SP)_1}{(SP)_2} = \frac{x + 0.04x}{x + 0.06x} = \frac{1.04x}{1.06x} = 52 : 53$

38. (b); Let quantity sold at loss be x kg and let CP per kg be Rs. 1. Total CP = Rs. 24

Total SP = Rs. [120% of (24 – x) + 95% of x]

$= \text{Rs.}\left[\frac{6}{5}(24 - x) + \frac{19x}{20}\right] = \text{Rs.}\left[\frac{576 - 5x}{20}\right]$

$\frac{576 - 5x}{20} = 110\% \text{ of } 24$, 576 – 5x = 528

5x = 48 $\Rightarrow$ x = Rs. 9.6

39. (a); Let Mark Price is Rs. 100

Selling Price = Rs. 90

Cost Price is $\frac{90}{1.5} = \text{Rs. } 60$

[on earning profit of 50%]

If discount is not given, Percentage Profit

will be $= \frac{100 - 60}{60} \times 100$ $\Rightarrow$ So, Profit = 66.67%

40. (a); If we are given 3 successive discounts of 15%, 10% and 5%,

So, the new reduced price after applying above discount on Rs. 10000

$= 10000 \times \frac{85}{100} \times \frac{90}{100} \times \frac{95}{100} = \text{Rs. } 7267.50$

Difficult

1. (c); Let us assume the cost price of the phone to be Rs. 100. Then,

$\text{Rs. } 100 \xrightarrow[\text{20\% profit}]{\text{sells at}} 100 \times 1.2 = \text{Rs. } 120 = (SP)_1$

If she bought the phone at 20% less, i.e. at Rs. 80 then,

$\text{Rs. } 80 \xrightarrow[\text{25\% profit}]{\text{sells at}} 80 \times 1.25 = \text{Rs. } 100 = (SP)_2$

So, $(SP)_1 - (SP)_2$ = Rs. (120 – 100)

= Rs. 20 when cost price is Rs. 100.

But, $(SP)_1 - (SP)_2$ = Rs. 180

$\therefore$ Cost price $= \frac{100}{20} \times 180 = \text{Rs. } 900$

Hence, the cost price of the book is Rs. 900

2. (a); Let the CP of each pen be Rs. 100

At the profit of 10%, SP of 40 pens

= (100 + 10) × 40 = Rs. 4400

At the profit of 20%, SP of 50 pens

= (100 + 20) × 50 = Rs. 6000

SP of 90 pens = Rs. (4400 + 6000) = Rs. 10400

CP of 90 pens = Rs. (90 × 100) = Rs. 9000
At the profit of 15%, SP of 90 pens
= Rs. (90 × 115) = Rs. 10350
Difference in SP = Rs. (10400 – 10350) = Rs. 50
If the difference is Rs. 50, then CP = Rs. 100
If the difference is Rs. 40, then CP

$$= \frac{100 \times 40}{50} = \text{Rs. } 80$$

Hence, the cost price of each pen is Rs. 80.

3. **(c);** Let Rs. p be the cost price of a shirt and Rs. q be the cost price of a pant. Then,
CP of 5 shirts = Rs. 5p
CP of 10 pants = Rs. 10q
$\therefore$ 5p + 10q = 1600 ...(i)

Profit on the sale of 5 shirts $= \frac{15 \times 5p}{100} = \text{Rs. } \frac{3p}{4}$

Loss on the sale of 10 pants $= \frac{10q \times 10}{100} = \text{Rs. } q$

Given,
Profit on the shirts – Loss on pants = Rs. 90

$$\Rightarrow \quad \frac{3p}{4} - q = 90$$

$\therefore$ 3p – 4q = 360 ...(ii)
Multiplying (i) by 3 and (ii) by 5 and then subtracting (ii) from (i), we get

$$50q = 3000 \quad \Rightarrow \quad q = \frac{3000}{50} = \text{Rs. } 60$$

Puting the value of q in (i) we get

$$5p = 1000 \quad \Rightarrow \quad p = \frac{1000}{5} = \text{Rs. } 200$$

Hence, the cost price of shirt is Rs. 200 each and the cost price of pant is Rs. 60 each.

4. **(a);** Total CP of articles = 750 × 0.6 = Rs. 450
[CP of 1 article = Rs. 0.6]
By selling 600 articles, Sarika should make a 40% profit on the outlay. This means that the selling price for 600 articles should be
1.4 × 450 = Rs. 630
Thus, selling price per article

$$= \frac{630}{600} = \frac{63}{60} = \text{Rs. } 1.05 \quad \text{[SP of 1 article]}$$

Since, Sarika sells only 630 articles at this price, her total recovery = 1.05 × 630 = Rs. 661.5
Hence, actual profit percent

$$= \frac{661.5 - 450}{450} \times 100 = 47\%$$

Thus, Sarika earns 47% profit on her total investment.

5. **(b);** Total number of i-phones = 15
$\therefore$ Total number of i-pads = 25 – 15 = 10
Total CP = Rs. 205000
Since, Kritika sells 80% of both goods at a profit of Rs. 40000, therefore, cost of 80% of the goods
= 0.8 × 205000 = Rs. 164000
Total amount recovered (or SP)
= Rs. (164000 + 40000) = Rs. 204000
Hence, loss = Rs. (205000 – 204000) = Rs. 1000
Hence, Kritika's overall loss is Rs. 1000

6. **(d);** Let the cost of sofa set be Rs. 100. Then, the cost of centre table is Rs. 40 [40% of sofa set given]
According to Sasha, cost of centre table

$$= \frac{90}{100} \times 40 = \text{Rs. } 36$$

[10% discount on center table]

and cost of sofa set $= \frac{75}{100} \times 100 = \text{Rs. } 75$

[25% discount on sofa set]

According to manager, He inter changed discount % so, cost of centre table

$$= \frac{75}{100} \times 40 = \text{Rs. } 30$$

and cost of sofa set $= \frac{90}{100} \times 100 = \text{Rs. } 90$

Extra money = Rs. (90 + 30) – Rs. (36 + 75)
= Rs. 120 – Rs. 111 = Rs. 9

$$\% \text{ extra} = \frac{9}{111} \times 100 = 8.1\%$$

Hence, Sasha paid 8.1% extra price.

7. **(c);** **Case 1:** If Rate is Rs. 3.25
Total sales revenue = 2000 × 3.25 × 0.75
= Rs. 4875
[Here 1000 strips are given free of cost]
Profit = Total sales revenue – Rs. 4800
= Rs. 4875 – Rs. 4800 = Rs. 75
Case 2: If rate is Rs. 4.25
Total sales revenue = 3000 × 4.25 × 0.75
= Rs. 9562.5 [Total 3000 strips]
Profit = Total sales revenue – Rs. 4800
= Rs. 9562.5 – Rs. 4800 = Rs. 4762.5
Hence, the ratio of profit is

$$\frac{\text{New profit}}{\text{Old profit}} = \frac{4762.5}{75} = 63.5$$

8. **(d);** Cost price = Rs. 240000 [Total 3000 copies]
Published price = Rs. 325 [Published price]

$$\text{Selling price} = \frac{75}{100} \times 325 = \text{Rs. } 243.75$$

No. of free copies $= 500 + \frac{2500}{25} = 500 + 100 = 600$

So, total selling price = 2400 × 243.75 = Rs. 585000

Hence, percentage gain $= \frac{585000 - 240000}{240000} \times 100$

$= \frac{345000}{240000} \times 100 = 143.75\%$

Hence, the overall gain is 143.75%.

9. (b); There were 5 i-phones (2 + 3) and 20 i-pads. Surbhi sells 2 i-phones for a profit of Rs. 2000 each. Hence, profit from i-phone sales = Rs. 4000

Then, profit from i-pads sales = Rs. 45000

Thus, profit per i-pad $= \frac{45000}{15} =$ Rs. 3000

(Since, 15 i-pads were sold in all.)

Hence, CP of i-pad = Rs. 15000

CP of i-phone = Rs. 7500

Total cost = Rs. 15000 × 20 + Rs. 7500 × 5
= Rs. 300000 + Rs. 37500 = Rs. 337500

[If only 75% of i-pads and 2 i-phones were sold]

Total revenue = Rs. 18000 × 15 + Rs. 9500 × 2
= Rs. 270000 + Rs. 19000 = Rs. 289000

Since, total revenue is less than total cost, there is a loss.

Hence, loss = Rs. 337500 – Rs. 289000 = Rs. 48500

Thus, Surbhi has an overall loss of Rs. 48500.

10. (b); In the given question, let the total profit% be p%

$\Rightarrow$ Total Profit p% $= \frac{1}{5} \times 5 + \frac{1}{4} \times 10 +$

$\frac{1}{2} \times 12 + \left[1 - \left(\frac{1}{5} + \frac{1}{4} + \frac{1}{2}\right)\right] \times 16$

$\Rightarrow 1 + \frac{5}{2} + 6 + \left(\frac{1}{20} \times 16\right) \Rightarrow \frac{103}{10}\%$

Now if he would have sold whole wheat at 11%, he would had made Rs. 72.80 more

$11\% - \frac{103}{10}\% =$ Rs. 72.80 , $\frac{7}{10}\% =$ Rs. 72.80

$1\% =$ Rs. $\frac{72.80 \times 10}{7}$ [Unitary method]

100% = Rs. 10,400 [Total CP of 4000 kg wheat]

CP of crop/kg = Rs. $\frac{10400}{4000} =$ Rs. 2.60

11. (d); Suppose selling price of both of them be Rs. P.

CP of Ajit = Rs. $p\left[\frac{100 - 25}{100}\right] =$ Rs. $\frac{3}{4}p$

(% Profit on SP)

CP of Rohit = Rs. $p\left[\frac{100}{100 + 25}\right] =$ Rs. $\frac{4}{5}p$

(% Profit on CP)

So, Ajit profit = SP – CP $= p - \frac{3}{4}p =$ Rs. $\frac{p}{4}$

Rohit profit = SP – CP $= p - \frac{4}{5}p =$ Rs. $\frac{p}{5}$

Difference of profit $= \frac{p}{4} - \frac{p}{5} =$ Rs. 100 (given)

$= \frac{p}{20} = 100 \Rightarrow$ SP = p = Rs. 2000

12. (d); Let CP of pen be Rs. x
and SP be Rs. y

Initially at loss of 20%,

$\Rightarrow \frac{20}{100} = \frac{x - y}{x} \Rightarrow \frac{1}{5}x = x - y$

$\Rightarrow y = \frac{4}{5}x$ ...(i)

Now if y would change to Rs. (y + 12), then profit becomes 30%

$\Rightarrow \frac{30}{100} = \frac{y + 12 - x}{x} \Rightarrow \frac{3}{10} = \frac{\frac{4}{5}x + 12 - x}{x}$

$\Rightarrow$ x = Rs. 24 ...(ii)

y = Rs. $\frac{96}{5}$ or Rs. 19.2 ...(iii)

%profit now if y becomes Rs. (y + 4.8)

$\Rightarrow$ %profit $= \frac{SP - CP}{CP} \times 100$

$= \frac{Rs.(y + 4.8) - 24}{24} \times 100$

$= \frac{Rs.(19.2 + 4.8) - 24}{24} \times 100 = 0\%$

13. (c); CP of i-phone to dealer
(on inclusion of 10% custom duty)
= Rs. (25000 + 10% of 25000) = Rs. 27500

The SP of i-phone at 20% profit

= Rs. $\left[27500 \times \frac{120}{100}\right] =$ Rs. 33000

But Here, Rs. 33000 is price after 25% discount on MP

Hence, MP × (1 – 0.25) = 33000

MP $= \frac{33000}{0.75} =$ Rs. 44000

14. **(a);** Let CP of cab driver be price of petrol

= Rs. 30 per liter

His, SP would be to carry 3 passengers

Let cost of 1 passenger be Rs. x

Initially he made profit of 20%

$$\Rightarrow p\% = \frac{20}{100} = \frac{SP - CP}{CP} = \frac{20}{100} = \frac{3x - 30}{30}$$

x = Rs. 12

Now, CP of petrol = Rs. 24 per litre

SP = 4 (cost of 1 passenger) = Rs. 48

$$= \text{Profit}\% = \frac{48 - 24}{24} \times 100 = 100\%$$

15. **(c);** Let the labelled price of article be Rs y.

$$\text{CP of article} = y\left(\frac{100 - 12.5}{100}\right) = \text{Rs.}\ \frac{7}{8}y$$

$$\text{SP of article} = y\left(\frac{100 + 17.5}{100}\right) = \text{Rs.}\ \frac{117.5}{100}\ y \text{ or}$$

$$= \text{Rs.}\ \frac{47}{40}y$$

$$\text{Profit} = \text{Rs.}\left[\frac{47}{40}y - \frac{7}{8}y\right] = \text{Rs.}\ \frac{12}{40}y$$

$$\%\text{profit} = \frac{\frac{12}{40}y}{\frac{7}{8}y} \times 100 = \frac{12}{35} \times 100 = 34\frac{2}{7}\%$$

16. **(b);** Let the labelled price of article be Rs. a

$$\text{CP of article} = \text{Rs. a}\left(\frac{100 - 15}{100}\right)$$

$$= \text{Rs.}\left[\frac{85}{100}\right]\text{a or Rs.}\ \frac{17}{20}\text{a}$$

$$\text{SP of article} = \text{Rs. a}\left(\frac{100 + 10}{100}\right)$$

$$= \text{Rs.}\left[\frac{110}{100}\right]\text{a or Rs.}\ \frac{11}{10}\text{a}$$

$$\text{Profit} = \text{Rs.}\left[\frac{11}{10}a - \frac{17}{20}a\right] = \text{Rs.}\ \frac{5a}{20}$$

$$\%\ \text{profit} = \frac{\frac{5a}{20}}{\frac{17a}{20}} \times 100 = \frac{5}{17} \times 100 = 29\frac{7}{17}\%$$

17. **(c);** Here, the labelled price of chair is

$$= \frac{SP \times 100}{(100 - D)\%} = \frac{2139 \times 100}{93} = \text{Rs. } 2300$$

Let the CP of chair be Rs. p

Now according to question:

$$\frac{15}{100} = \frac{2300 - p}{p} \Rightarrow 15p = 230000 - 100p$$

p = Rs. 2000

18. **(c);** Here, the labelled price of decks is defined as:

$$\Rightarrow \frac{SP \times 100}{(100 - D)\%} = \frac{166 \times 100}{100 - 17} = \text{Rs. } 200$$

Let CP of deck be p

According to the question:

$$\frac{25}{100} = \frac{200 - p}{p} \Rightarrow P = \text{Rs. } 160$$

19. **(c);** The original company price

$$= \text{Rs.}\left[\frac{25000 \times 100}{100 - 15}\right] = \text{Rs.}\left[\frac{25000 \times 100}{85}\right]$$

= Rs. 29,411.76

Let the total SP be Rs. p

Now according to question:

$$= \frac{p - 29411.76}{29411.76} \times 100 = 8 \quad \Rightarrow \quad p = \text{Rs. } 31{,}764.70$$

20. **(a);** The original company price $= \text{Rs.}\left[\frac{SP \times 100}{(100 - D)\%}\right]$

$$= \text{Rs.}\left[\frac{860 \times 100}{(100 - 14)\%}\right] = \text{Rs. } 1000$$

Let the total SP be Rs. q

Now according to equation:

$$= \frac{SP - CP}{CP} \times 100 = (\text{profit})\%$$

$$\Rightarrow \frac{q - 1000}{1000} \times 100 = 6 \quad \Rightarrow q = \text{Rs. } 1060$$

Previous Year (Memory Based)

1. **(b);** $CP = \frac{8}{10} = 0.8$

SP = 1.25

$$\text{gain\%} = \frac{1.25-0.8}{0.8}\times 100 = \frac{225}{4} = 56\frac{1}{4}\%$$

2. **(a);** Let CP = x Rs.

$\therefore$ 515 – x = x – 475

2x = 990 $\Rightarrow$ x = 495

$\therefore$ Required SP $= \frac{140}{100}\times 495 = 693$ Rs.

3. **(a);** $+1 \text{ unit}\left(\begin{array}{c}10\\ \\ 9\end{array}\right) - 10\%$

$$\frac{\text{Reduced price}}{\text{Original price}} = \frac{9}{10}$$

$\therefore$ Required Reduced price $= \frac{225}{25\times 10} =$ Rs. 0.9

Required original price $= \frac{225}{25\times 9} =$ Rs. 1

4. **(c);** $-1 \text{ unit}\left(\begin{array}{c}4\\ \\ 5\end{array}\right) + 25\%$

$$\frac{\text{increased price}}{\text{original price}} = \frac{5}{4}$$

$\therefore$ Required original price $= \frac{75}{2\times 5} = 7.5$ Rs.

5. **(b);** $CP = \frac{15}{12} = \frac{5}{4} \Rightarrow SP = \frac{14}{10} = \frac{7}{5}$

$$\therefore \text{Required profit\%} = \frac{\frac{7}{5}-\frac{5}{4}}{\frac{5}{4}}\times 100$$

$$= \frac{3}{25}\times 100 = 12\%$$

6. **(c);** Let CP = 100

actual CP = 70

SP = 100 – 44 = 56

$$\therefore \text{loss\%} = \frac{70-56}{70}\times 100 = \frac{14}{70}\times 100 = 20\%$$

7. **(c);** Let CP = 100

Actual CP = 125

SP = 100 + 25 = 125

$\therefore$ Neither loss nor gain

8. **(d);** CP : SP

100 : 114

100 : 117

$\therefore$ 3 units = 15

100 units = 500 Rs.

9. **(b);** CP : SP

100 : 92

100 : 107

$\therefore$ 15 units = 120

100 units $= \frac{120}{15}\times 100 = 800$ Rs.

10. **(a);** Let CP 100x $\Rightarrow$ (100x – 56) New CP

SP 115x $\Rightarrow$ (115x – 56) New SP

$$\frac{122}{100}(100x-56) = 115x-56$$

122x – 1.22 × 56 = 115x – 56

x = 1.76

$\therefore$ Required CP = 1.76 × 100 = 176 Rs.

11. **(b);** Let CP = 100x $\Rightarrow$ 100x – 25 New CP

SP = 130x $\Rightarrow$ 130x – 19 New SP

$$\frac{150}{100}(100x-25) = 130x-19$$

$$150x - \frac{150}{4} = 130x - 19$$

$$20x = \frac{150}{4} - 19 \Rightarrow x = 0.925$$

$\therefore$ Required CP = 92.5 Rs.

12. **(b);** Let CP = 100x $\Rightarrow$ 100x + 80 New CP

SP = 145x = 145x + 70 New SP

$$\therefore \frac{130}{100}(100x+80) = 145x+70$$

$$130x + \frac{130\times 80}{100} = 145x+70$$

15x = 13 × 8 – 7

x = 2.2666

$\therefore$ Required CP $= 226\frac{2}{3}$ Rs.

13. (d); Let CP is Rs. x

According to question

$$\frac{1}{5}\times\frac{25}{100}\times x+\frac{1}{10}\times x\times\frac{40}{100}+\frac{7}{10}\times x\times\frac{30}{100}=150$$

$$\frac{5x}{100}+\frac{4x}{100}+\frac{21x}{100}=150$$

$$=\frac{150\times100}{30}=500$$

14. (a); Let CP of horse sold at gain = A

CP of horse sold at loss = B

160% of A = 80% of B $\Rightarrow \frac{A}{B}=\frac{1}{2}$

CP of A = $\frac{1}{3}\times960=320$

CP of B $=\frac{2}{3}\times960=640$

15. (a); Let value of consignment = 800

$\frac{5}{8}$ of consignment = 500

$\text{loss}=\frac{16}{100}\times500=80$ Rs.

rest value = 800 – 500 = 300 Rs.

$\text{profit}=\frac{8}{100}\times300=24$ Rs.

$\therefore$ Overall loss = 80 – 24 = 56 Rs.

$\therefore 56\rightarrow140$

$1\rightarrow\frac{140}{56}$

$800\rightarrow\frac{140}{56}\times800=2000$ Rs.

16. (a); Let Required% = x

-15% $\qquad$ $+x\%$

$+10\%$

$(x-10)\%$ $\qquad$ $(10+15)=25\%$

Given $(x-10):25::2:1$

$\therefore x-10=50$

$x=60\%$

17. (c); Let Required% = x

$+24\%$ $\qquad$ $X\%$

21%

$(21-x)\%$ $\qquad$ 3%

Given, $(21-x):3=3:5$

$\therefore x=19.2$

$x=19\frac{1}{5}\%$

18. (a); Let oranges bought = 240

$\therefore$ Total CP $=\frac{120}{12}+\frac{120}{8}=10+15=25$ Rs.

$SP=\frac{240}{15}=16$ Rs.

$\therefore \text{loss}\%=\frac{25-16}{25}\times100=36\%$

19. (a); Let CP of two tables be A and B

110% A = 105% of B $\Rightarrow \frac{A}{B}=\frac{21}{22}$

CP of A = $\frac{21}{43}\times860=420$

CP of B = $\frac{22}{43}\times860=440$

20. (c); Total profit $=\frac{120}{1000}\times100=12\%$

Pen $\qquad$ Pencils

15% $\qquad$ 10%

12%

2 $\qquad$ 3

Total Cost price of Pen $=\frac{2}{5}\times1000=400$ Rs.

Total Cost price of Pencils $=\frac{3}{5}\times1000=600$ Rs.

$\therefore$ Cost of a Pen $=\frac{400}{5}=80$ Rs.

Cost of a Pencil $=\frac{600}{30}=20$ Rs.

21. (c); Let no. of oranges bought = 240

$$\therefore \text{Total CP} = \frac{120}{12} + \frac{120}{80} = 10 + 15 = 25 \text{ Rs.}$$

$$\text{Total SP} = \frac{240}{10} = 24 \text{ Rs.}$$

loss = 25 – 24 = 1 Rs.

$\therefore 1 \rightarrow 8$

$240 \rightarrow 8 \times 240 = 1920$

22. (d);

1st   2nd

– 23%   27%

0%

27 : 23

$$\therefore \text{Cost price of 1st horse} = \frac{27}{50} \times 1550 = 837 \text{ Rs.}$$

$$\text{Cost price of 2nd horse} = \frac{23}{50} \times 1550 = 713 \text{ Rs.}$$

23. (c); Let original price = 100

$$SP = \frac{110}{100} \times 95 = 104.5$$

$\therefore$ Required % profit = 4.5%

24. (a);

$$\frac{\text{Patato}}{\text{Tomato}} = \frac{5}{20} = \frac{1}{4}$$

$$\frac{\text{Tomato}}{\text{Onion}} = \frac{30}{12} = \frac{5}{2}$$

$$\frac{\text{Onion}}{\text{Cabbage}} = \frac{18}{15} = \frac{6}{5}$$

$$1 \text{ kg of Cabbage} = \frac{50}{10} = 5 \text{ Rs./kg.}$$

$\therefore$ Potato : Tomato : Onion : Cabbage

15 : 60 : 24 : 20

$\therefore$ Potato : Cabbage = 3 : 4

$$\therefore \text{Required Price} = \frac{5}{4} \times 3 \times 24 = 90 \text{ Rs.}$$

25. (a);

	1st	:	2nd		:	
Cost price	10	:	10	110	:	120
SP	$12_{\times 11}$	:	$11_{\times 12}$	132	:	132

Given, SP = 640 Rs.

total SP = 1280 Rs.

$132 \rightarrow 640$

$$110 \rightarrow \frac{640}{132} \times 110 = 533.33$$

$$120 \rightarrow \frac{640}{132} \times 120 = 581.8181$$

Total CP = 1115.148

$$\therefore \text{Required profit\%} = \frac{164.85}{1115.14} \times 100 = 14.78\%$$

26. (a);

	Miheer	Satya		
CP	90	20	1890	2000
SP	$100_{\times 21}$	$21_{\times 100}$	2100	2100

difference of their profit = 110

So, Selling price = 2100

27. (b);

	First	:	Second
CP	100	:	80
SP	70	:	100

Given total SP = 11900

$170 \rightarrow 11900$

$$1 \rightarrow \frac{11900}{170}$$

$$180 \rightarrow \frac{11900}{170} \times 180$$

Total CP = 12600

Loss = 12600 – 11900 = 700 Rs.

28. (b);

+ 35%   – 30%

O

30 : 35

Ratio = 6 : 7

$$\text{CP of first table} = \frac{100}{135} \times 405 = 300 \text{ Rs.}$$

$$\therefore \text{CP of second table} = \frac{300}{6} \times 7 = 350 \text{ Rs.}$$

29. (d); Let CP = 100, SP = x

$$25 = \frac{100 - \frac{x}{2}}{100} \times 100$$

$$25 = 100 - \frac{x}{2}, \quad \frac{x}{2} = 75$$

x = 150 Rs.

$$\text{Required \% profit} = \frac{50}{100} \times 100 = 50\%$$

30. (c); CP LP SP

70 100 112

Required% $= \frac{42}{70} \times 100 = 60\%$

31. (b); CP MP SP

200 250 210

Required% $= \frac{50}{200} \times 100 = 25\%$

32. (d); Total CP = 25 × 6 + 35 × 7

= 150 + 245 = 395 Rs.

SP = 60 × 6.75 = 405 Rs.

gain = 10 Rs.

33. (c); Required price $= \frac{100}{140} \times \frac{100}{120} \times 10500 = 6250$ Rs.

34. (b); CP $= \frac{100}{133} \times 4921 = 3700$ Rs.

35. (d); Required SP $= \frac{130}{100}(950 + 300) = 1625$ Rs.

36. (d); Total CP when discount not offered

$= \frac{100}{90} \times 11250 + 150 + 800 = 13450$ Rs.

Required SP $= \frac{115}{100} \times 13450 = 15467.5$ Rs.

37. (d); CP $= \frac{100}{122} \times 10980 = 9000$ Rs.

38. (a); Required amount $= \frac{95}{100} \times (25000) \times 12$

$= 285000$ Rs.

39. (c); $100 \xrightarrow{-5\%} 95 \xrightarrow{+10\%} 104.5$

Required% = 4.5%

40. (a); original price of perfume $= \frac{100}{85} \times 3675.4 = 4324$